IOT BASED AIR QUALITY MONITORING DEVICE

22TP708 - EXPERIMENTAL PROJECT BASED LEARNING

Submitted by

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

The Air pollution has become one of the most serious environmental and health challenges in modern society. Rapid urbanization, industrial activities, and increasing vehicle emissions have led to a significant rise in harmful pollutants in the air. Continuous exposure to polluted air can cause respiratory problems, allergies, and long-term health issues. However, most individuals lack access to real-time, localized air quality information, which makes it difficult to take preventive actions in daily life.

This project proposes a wearable air quality monitoring watch that enables users to measure and observe air pollution levels in their immediate surroundings. The system is designed to be compact, portable, and user-friendly, allowing continuous monitoring without the need for external devices. It utilizes an MQ-135 gas sensor to detect harmful gases such as carbon dioxide, ammonia, and smoke present in the environment. The sensor data is processed using a Node MCU ESP8266 microcontroller, which converts the readings into meaningful air quality values.

The processed data is displayed directly on an OLED screen attached to the wearable device. The system shows real-time air quality levels in terms of PPM (Parts Per Million) along with simple indicators such as good, moderate, or poor air quality. In addition, an alert mechanism such as a buzzer or vibration is integrated into the system to notify users when pollution levels exceed safe limits. This immediate feedback helps users take quick actions, such as avoiding certain areas or using protective measures.

The proposed system offers several advantages, including real-time monitoring, portability, low cost, and ease of use. It does not rely on internet connectivity or mobile applications, making it more accessible and reliable in various conditions. The device can be used by students, office workers, and the general public to stay informed about environmental conditions and protect their health.

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CHAPTER 1

INTRODUCTION

Air pollution has become one of the most critical environmental challenges in recent years, significantly affecting human health, climate, and overall quality of life. Rapid industrialization, urbanization, and the increasing number of vehicles have led to a drastic rise in harmful gases such as carbon dioxide (CO_2), ammonia (NH_3), nitrogen oxides (NOx), and other pollutants in the atmosphere. Continuous exposure to such pollutants can cause severe health problems, including respiratory diseases, allergies, and long-term chronic conditions. Despite the seriousness of this issue, access to real-time and location-specific air quality information is still limited for the general public. Most traditional air quality monitoring systems are expensive, complex, and installed only in specific locations, making them inaccessible for personal or small-scale use.

To address these challenges, this project presents the design and development of an IoT-based air quality monitoring system using the ESP8266 microcontroller and MQ-135 gas sensor. The proposed system aims to provide a low-cost, efficient, and real-time solution for monitoring air quality in both indoor and outdoor environments. The MQ-135 sensor is capable of detecting various harmful gases present in the air, and it continuously sends analog data to the ESP8266 microcontroller. The ESP8266 acts as the central processing unit, which processes the sensor data and converts it into meaningful air quality information. One of the key features of the ESP8266 is its built-in WiFi capability, which enables seamless communication and data transmission over the internet.

The system is designed to host a web server using the IP address generated by the ESP8266. Users can access this IP address through any web browser on devices such as laptops, smartphones, or tablets to view real-time air quality data. This web-based interface provides a simple and user-friendly way to monitor environmental conditions without requiring any additional applications or software installations. The real-time monitoring capability ensures that users are always aware of the air quality in their surroundings, enabling them to take necessary precautions when pollution levels rise.

In addition to monitoring, the system also incorporates an alert mechanism using a buzzer. When the detected air quality exceeds a predefined threshold level, the buzzer is activated to provide an immediate warning to the user. This feature enhances the safety aspect of the system by ensuring that users are promptly notified of hazardous conditions. The threshold levels can be adjusted based on the requirements of the user or the environment in which the system is deployed.

The proposed system offers several advantages over existing air quality monitoring methods. It is cost-effective, portable, easy to install, and requires minimal maintenance. Unlike traditional monitoring stations, this system provides localized and real-time data, making it more relevant for individual users. Furthermore, the integration of IoT technology enables remote access and continuous monitoring, which are essential features in modern smart environments. The simplicity of the design also makes it suitable for educational purposes and small-scale implementations.

However, the system also has certain limitations, such as the need for proper calibration of the MQ135 sensor to ensure accurate readings and the dependence on a stable WiFi connection for data transmission. Environmental factors such as temperature and humidity may also influence sensor performance. Despite these challenges, the system serves as an effective prototype for demonstrating the capabilities of IoT in environmental monitoring.

1.1 BACKGROUND

Air pollution is one of the most serious environmental problems faced by the world today. It refers to the presence of harmful substances in the atmosphere that can adversely affect human health, animals, plants, and the overall ecosystem. Over the past few decades, rapid industrialization, urbanization, and population growth have significantly contributed to the increase in air pollution levels. The emission of harmful gases from vehicles, factories, and other human activities has led to the deterioration of air quality in both urban and rural areas. Pollutants such as carbon monoxide (CO), carbon dioxide (CO_2), sulfur dioxide (SO_2), nitrogen oxides (NOx), and particulate matter (PM) are commonly found in polluted air, and prolonged exposure to these substances can lead to severe health complications.

The impact of air pollution on human health is a major concern. It is directly linked to respiratory diseases such as asthma, bronchitis, and lung infections. In addition, long-term exposure to polluted air can cause cardiovascular diseases and even cancer. Children, elderly individuals, and people with pre-existing health conditions are particularly vulnerable to the effects of poor air quality. According to various environmental studies, millions of deaths occur every year due to air pollution-related illnesses. This highlights the urgent need for effective monitoring and control of air pollution to protect public health and improve the quality of life.

Traditional air quality monitoring systems are generally implemented by government organizations and environmental agencies. These systems consist of large, sophisticated equipment that can accurately measure different types of pollutants in the atmosphere. While these systems provide reliable data, they are expensive to install and maintain. Moreover, they are usually deployed in limited locations, such as major cities or industrial areas, which makes it difficult to obtain localized air quality information for specific regions or individual households. As a result, people often lack access to real-time data about the air they breathe in their immediate surroundings.

With the advancement of technology, there has been a growing interest in developing low-cost and portable air quality monitoring systems. The emergence of microcontrollers and sensors has made it possible to design compact devices that can measure environmental parameters effectively. Sensors such as the MQ-series gas sensors are widely used for detecting various gases present in the air. Among them, the MQ-135 sensor is particularly suitable for air quality monitoring, as it can detect gases like ammonia, benzene, smoke, and carbon dioxide. These sensors convert the concentration of gases into electrical signals, which can then be processed by microcontrollers.

The development of the Internet of Things (IoT) has further revolutionized the field of environmental monitoring. IoT refers to the network of interconnected devices that can communicate and exchange data over the internet. By integrating sensors with IoT-enabled microcontrollers, it is possible to create smart systems that can monitor environmental conditions in real time and provide remote access to data. The ESP8266 microcontroller is one such device that has gained popularity due to its built-in WiFi capability, low cost, and ease of use. It allows developers to create wireless applications where data can be transmitted and accessed through the internet.

In modern IoT-based air quality monitoring systems, data collected from sensors is processed and displayed using web-based interfaces or mobile applications. This enables users to monitor air quality from anywhere at any time. The use of web servers hosted on microcontrollers allows real-time data visualization without the need for complex infrastructure. Additionally, alert mechanisms such as buzzers or notifications can be incorporated to warn users when pollution levels exceed safe limits.
These features make IoT-based systems more efficient and user-friendly compared to traditional monitoring methods.

Despite the advantages of IoT-based monitoring systems, there are still certain challenges that need to be addressed. One of the main issues is the accuracy of low-cost sensors, which may require proper calibration to provide reliable results. Environmental factors such as temperature and humidity can also affect sensor performance. Furthermore, the dependency on internet connectivity may limit the functionality of the system in areas with poor network coverage. Security and privacy concerns related to data transmission are also important considerations in IoT applications.

The need for real-time and accessible air quality monitoring has become more important than ever, especially in densely populated and industrialized regions. Individuals require accurate and up-to-date information about the air they breathe to make informed decisions and take preventive measures. This has led to the development of innovative solutions that combine sensing technology, wireless communication, and data visualization. These systems aim to provide affordable and efficient monitoring solutions that can be used by a wide range of users.

In this context, the proposed IoT-based air quality monitoring device plays a significant role. By utilizing the MQ-135 sensor and ESP8266 microcontroller, the system is capable of detecting air pollution levels and displaying the data in real time through a web interface. The inclusion of a buzzer for alerts further enhances the functionality of the system by providing immediate warnings in case of hazardous conditions. The simplicity and cost-effectiveness of the design make it suitable for both educational and practical applications.

Overall, the background study emphasizes the importance of air quality monitoring and the need for modern technological solutions to address environmental challenges. It highlights the transition from traditional monitoring systems to smart IoT-based systems that offer real-time data, remote accessibility, and user-friendly interfaces. The proposed project is a step towards creating a smarter and healthier environment by enabling individuals to monitor and respond to air pollution effectively.

1.2 MOTIVATION

Air pollution has emerged as a major environmental and public health concern across the globe. With the rapid growth of industries, urban development, and an increase in vehicular emissions, the quality of air has deteriorated significantly over the years. In many cities and towns, people are exposed to harmful gases and pollutants on a daily basis without being aware of the actual air quality around them. This lack of awareness and accessibility to real-time air quality data has become one of the key motivating factors behind the development of this project. The need to monitor air pollution at a personal and local level has inspired the creation of a system that is both affordable and easy to use.

One of the primary motivations for this project is the impact of air pollution on human health. Continuous exposure to polluted air can lead to severe respiratory problems, allergies, asthma, and other chronic diseases. Vulnerable groups such as children, elderly individuals, and people with preexisting health conditions are at a higher risk. Despite these serious health implications, most people do not have access to real-time information about the air they breathe in their homes, workplaces, or surroundings. This gap between environmental conditions and public awareness highlights the need for a system that can provide immediate and accurate information about air quality, enabling individuals to take preventive measures.

Another important motivation is the limitation of existing air quality monitoring systems. Traditional monitoring stations are expensive, complex, and installed only in selected locations. These systems are not designed for personal use and cannot provide localized data for specific areas such as homes, classrooms, or small offices. Even though mobile applications and online platforms provide air quality information, they often rely on data from distant monitoring stations, which may not accurately represent the conditions in a user’s immediate environment. This limitation emphasizes the importance of developing a compact and portable solution that can measure air quality directly at the user’s location.

The advancement of IoT (Internet of Things) technology has also played a significant role in motivating this project. IoT enables devices to communicate with each other and share data over the internet, making it possible to monitor environmental conditions in real time. The availability of lowcost microcontrollers like ESP8266, along with sensors such as MQ-135, has made it feasible to design smart systems that are both efficient and economical. This technological progress has encouraged the development of innovative solutions that can address real-world problems in a simple and effective manner. The integration of IoT in air quality monitoring allows users to access data remotely through web interfaces, enhancing convenience and usability.

Cost-effectiveness is another key factor that motivated the development of this project. Many advanced monitoring systems are not affordable for individuals or small organizations. There is a growing need for a low-cost alternative that can deliver reliable performance without requiring significant investment. By using readily available components such as ESP8266, MQ-135 sensor, and a simple buzzer, this project aims to provide an economical solution that can be easily implemented by students, researchers, and general users. This makes the system suitable not only for practical applications but also for educational purposes.

The idea of creating a user-friendly system has also been a strong motivation. Many existing systems require technical expertise to operate and maintain, which limits their usability among non-technical users. This project focuses on simplicity and ease of use by providing a web-based interface that can be accessed through any standard browser. Users can simply enter the IP address of the device to view real-time air quality data, eliminating the need for additional software or applications. This approach ensures that the system is accessible to a wider audience, including individuals with minimal technical knowledge.

Environmental awareness and sustainability are also important motivating factors. As the world faces increasing environmental challenges, there is a need to promote awareness and encourage responsible behavior among individuals. By providing real-time information about air quality, this system helps users understand the level of pollution in their surroundings and motivates them to take necessary actions, such as improving ventilation, reducing exposure, or avoiding polluted areas. This contributes to a healthier lifestyle and supports efforts towards environmental conservation.

In addition, the integration of an alert system using a buzzer adds a practical dimension to the project. The buzzer provides immediate notification when pollution levels exceed safe limits, ensuring that users are promptly informed about hazardous conditions. This feature enhances the safety and reliability of the system, making it more effective in real-world

applications. The ability to receive instant alerts without constantly monitoring the web interface is a significant advantage, especially in critical situations.
Finally, the motivation for this project is also driven by the potential for future improvements and innovations. The system can be further enhanced by integrating additional sensors, cloud storage, mobile applications, and advanced data analysis techniques. This opens up opportunities for research and development in the field of smart environmental monitoring. The project serves as a foundation for exploring more advanced solutions that can contribute to the development of smart cities and intelligent systems.

1.3 OBJECTIVES

The primary objective of this project is to design and develop an efficient, low-cost, and real-time air quality monitoring system using IoT technology. Air pollution has become a serious concern in modern society, and there is an increasing need for systems that can continuously monitor environmental conditions and provide accurate information to users. This project aims to utilize the ESP8266 microcontroller and MQ-135 gas sensor to detect harmful gases in the air and present the data in a user-friendly manner. The system is intended to bridge the gap between complex traditional monitoring systems and the need for simple, accessible solutions for everyday use.

One of the key objectives of this project is to measure air quality in real time. The MQ-135 sensor plays a vital role in detecting various harmful gases such as carbon dioxide, ammonia, and smoke present in the atmosphere. The sensor converts these gas concentrations into electrical signals, which are then processed by the ESP8266 microcontroller. By continuously collecting and analyzing this data, the system ensures that users receive up-to-date information about air quality. This real-time monitoring capability is essential for identifying sudden changes in environmental conditions and taking immediate action.

Another important objective is to provide remote accessibility of air quality data. The ESP8266 microcontroller is equipped with built-in WiFi functionality, which allows it to connect to a local network and host a web server. The system generates an IP address that can be accessed through any web browser on devices such as smartphones, laptops, or tablets. This enables users to monitor air quality from anywhere within the network without the need for specialized applications. The webbased interface is designed to be simple and easy to understand, ensuring that even non-technical users can access and interpret the data effectively.

The project also aims to implement an alert mechanism to enhance user safety. A buzzer is integrated into the system to provide immediate notification when air quality exceeds predefined safe limits. This feature ensures that users are alerted to potentially hazardous conditions without the need to constantly monitor the display. The threshold levels for triggering the buzzer can be set based on user requirements or environmental standards. This objective focuses on making the system not only informative but also proactive in protecting user health.

Cost-effectiveness and simplicity are also major objectives of this project. Many existing air quality monitoring systems are expensive and require complex installation and maintenance procedures. This project aims to develop a solution that is affordable and easy to implement using readily available components such as ESP8266, MQ-135 sensor, and basic electronic components. By minimizing the cost and complexity, the system becomes accessible to a wider range of users, including students, researchers, and small organizations. This objective highlights the importance of creating practical solutions that can be widely adopted.

Another objective is to ensure portability and flexibility of the system. The compact design of the components allows the device to be easily installed in different environments, such as homes, offices, classrooms, and industrial areas. Unlike traditional monitoring systems that are fixed in specific locations, this system can be relocated based on user requirements. This flexibility makes it suitable for a variety of applications and enhances its usability in different scenarios.

The project also focuses on promoting environmental awareness among users. By providing real-time information about air quality, the system encourages individuals to be more conscious of their surroundings and take necessary precautions to reduce exposure to pollution. It also helps users understand the impact of their activities on the environment and motivates them to adopt more sustainable practices. This objective emphasizes the role of technology in educating and empowering individuals to contribute to environmental protection.

Reliability and efficiency are additional objectives of the system. The project aims to ensure that the device operates consistently and provides accurate data under normal conditions. Although low-cost sensors may have certain limitations, efforts are made to optimize their performance through proper calibration and programming. The system is designed to function continuously with minimal interruptions, ensuring that users can rely on it for consistent monitoring.

Furthermore, the project aims to create a foundation for future enhancements and research. The system can be extended by integrating additional sensors to detect more environmental parameters such as temperature, humidity, and particulate matter. It can also be connected to cloud platforms for data storage and advanced analysis. These possibilities open up new opportunities for innovation and development in the field of smart environmental monitoring. This objective highlights the scalability and adaptability of the system.

1.4 SCOPE OF THE PROJECT

The scope of this project focuses on the design and development of a low-cost, real-time air quality monitoring system using IoT technology. The system is intended to measure the concentration of harmful gases present in the environment using the MQ-135 gas sensor and process the data using the ESP8266 microcontroller. The primary aim is to provide users with real-time information about air quality through a web-based interface that can be accessed using the IP address generated by the device. This project is designed to be simple, efficient, and suitable for both indoor and outdoor applications.

One of the major aspects of the project scope is real-time monitoring of air quality. The system continuously senses environmental conditions and provides instant updates on pollution levels. This enables users to stay informed about the quality of air in their surroundings at all times. The real-time capability ensures that sudden changes in air quality can be detected immediately, allowing users to take necessary precautions. This feature is particularly useful in environments where pollution levels can fluctuate rapidly, such as industrial areas, traffic zones, and enclosed indoor spaces.

Another important aspect of the scope is the use of wireless communication through the ESP8266 microcontroller. The built-in WiFi functionality allows the system to connect to a local network and host a web server. Users can access the air quality data by entering the device's IP address into a web browser. This eliminates the need for additional software or applications, making the system more user-friendly and accessible. The use of web-based monitoring also enables multiple users to access the data simultaneously, enhancing the practicality of the system.

The project also includes an alert mechanism as part of its scope. A buzzer is integrated into the system to provide immediate notification when the air quality exceeds a predefined threshold level. This ensures that users are alerted to potentially harmful conditions without the need to continuously monitor the display. The alert system enhances the safety and reliability of the device, making it suitable for use in environments where quick response is necessary.

Portability and ease of installation are also key elements within the scope of this project. The system is designed using compact and lightweight components, allowing it to be easily installed and relocated as needed. Unlike traditional air quality monitoring systems that are fixed in specific locations, this device can be used in various environments such as homes, offices, schools, and small industries. This flexibility makes the system highly adaptable to different use cases and user requirements.

The scope of the project also extends to cost-effectiveness. By using affordable components such as the ESP8266 and MQ-135 sensor, the system provides a budget-friendly solution for air quality monitoring. This makes it accessible to a wide range of users, including students, researchers, and individuals who require a simple monitoring system for personal use. The low cost does not compromise the functionality of the system, as it still provides essential features such as real-time monitoring, remote access, and alert notifications.

Another important aspect of the project scope is its educational value. The system serves as a practical example of how IoT technology can be applied to solve real-world problems. It helps students and beginners understand the integration of sensors, microcontrollers, and wireless communication in a simple and effective manner. The project can be used as a learning tool in academic institutions to demonstrate concepts related to embedded systems, IoT, and environmental monitoring.

The scope of the project can be further extended in the future by incorporating additional features and technologies. For example, more advanced sensors can be added to measure parameters such as temperature, humidity, and particulate matter. The system can also be integrated with cloud platforms for data storage and analysis, enabling long-term monitoring and trend analysis. Mobile applications can be developed to provide notifications and enhance user interaction. These potential enhancements highlight the scalability of the system and its ability to evolve with technological advancements.

CHAPTER II

LITERATURE SURVEY

2.1 LITERATURE REVIEW

[1] AUTHOR(S): IEEE Access

YEAR: 2024

TITLE: “IoT-Based Smart Air Quality Monitoring System Using ESP8266”

Description:

This paper presents the design and development of a smart air quality monitoring system using IoT technology and the ESP8266 microcontroller. The system focuses on real-time monitoring of environmental conditions by integrating gas sensors capable of detecting harmful pollutants in the air. The collected data is processed by the ESP8266 and transmitted over WiFi to a web-based platform for visualization. The study emphasizes the importance of continuous monitoring in urban areas where pollution levels fluctuate rapidly. The system is designed to be low-cost and user-friendly, making it suitable for both indoor and outdoor environments. The authors highlight the advantages of using ESP8266 due to its built-in WiFi capability and ease of implementation. The system also supports remote access, allowing users to monitor air quality from anywhere. Experimental results show that the system performs efficiently in detecting pollution levels and updating data in real time. The paper concludes that IoT-based monitoring systems can significantly improve environmental awareness and support smart city initiatives.

[2] AUTHOR(S): Elsevier

YEAR: 2024

TITLE: “Real-Time Air Quality Monitoring Using Low-Cost Sensors and IoT”

Description:

This research focuses on developing an affordable air quality monitoring system using low-cost sensors integrated with IoT technology. The system continuously measures pollutants in the air and transmits the data to a centralized platform for analysis and visualization. The authors emphasize the need for cost-effective solutions to address air pollution monitoring in developing regions. The system architecture includes sensors, a microcontroller, and a communication module that enables real-time data transfer. The study highlights the benefits of using IoT for continuous monitoring and remote accessibility. It also discusses the importance of data collection for environmental analysis and decision-making. The results demonstrate that low-cost systems can provide acceptable performance for general monitoring purposes. The system is easy to deploy and can be used in various environments such as homes, offices, and industrial areas.

[9] AUTHOR(S): Elsevier YEAR: 2024

TITLE: "Multi-Gas Monitoring System Using IoT"

Description:

This paper discusses the development of a multi-gas monitoring system using IoT technology. The system uses multiple sensors to detect different gases in the environment. The collected data is processed and transmitted for analysis. The authors highlight the importance of detecting multiple pollutants for accurate environmental monitoring. The system is suitable for industrial and environmental safety applications.

[3] AUTHOR(S): IEEE Sensors Journal

YEAR: 2025

TITLE: "Wireless Sensor Network-Based Air Quality Monitoring"

Description:

This study presents a wireless sensor network (WSN)-based approach for monitoring air quality. Multiple sensor nodes are deployed in different locations to collect environmental data, which is then transmitted to a central server for processing and analysis. The system improves coverage and provides more accurate data by using multiple sensors. The authors emphasize the importance of distributed monitoring systems in large-scale environments such as cities and industrial areas. The use of wireless communication enables efficient data transfer without the need for wired connections. The system also supports real-time monitoring and data visualization.

[4] AUTHOR(S): Elsevier

YEAR: 2024

TITLE: "Indoor Air Quality Monitoring Using IoT"

Description:

This paper focuses on monitoring indoor air quality using IoT-based systems. It highlights the importance of maintaining healthy indoor environments, especially in homes, offices, and educational institutions. The system uses sensors to detect pollutants such as carbon dioxide and volatile organic compounds (VOCs). The collected data is processed and displayed through a web interface or mobile application. The study emphasizes the need for real-time monitoring to ensure safe indoor conditions.

The system is compact, easy to install, and suitable for indoor applications.

[5]AUTHOR(S): IEEE Internet of Things Journal

YEAR: 2025

TITLE: "IoT and AI-Based Air Quality Prediction System"

Description:

This research combines IoT technology with artificial intelligence to develop an advanced air quality monitoring and prediction system. The system collects real-time data using sensors and applies machine learning algorithms to predict future pollution levels. The authors highlight the importance of predictive analysis in preventing environmental hazards. The system provides both real-time monitoring and future predictions, making it highly useful for decision-making and planning. The integration of AI enhances the accuracy and efficiency of the system

[6] AUTHOR(S): Springer

YEAR: 2025

TITLE: "Smart Environmental Monitoring Using IoT"

Description:

This paper presents a comprehensive environmental monitoring system using IoT technology. The system integrates multiple sensors to monitor various parameters such as air quality, temperature, and humidity. The collected data is processed and displayeDthrough a centralized platform. The authors emphasize the importance of monitoring multiple environmental factors for better analysis. The system is designed for smart city applications and supports real-time data access.

2.2 SUMMARY OF LITERATURE REVIEW

The literature review on IoT-based air quality monitoring systems highlights the growing importance of monitoring environmental pollution using modern technological solutions. Various research works from recent years demonstrate that air pollution has become a critical issue affecting human health and the environment. Traditional air quality monitoring systems, although accurate, are expensive, complex, and limited to specific locations. This has led researchers to explore alternative approaches that are cost-effective, portable, and capable of providing real-time data. The integration of IoT technology with low-cost sensors has emerged as a promising solution to overcome these limitations.

AUTHOR(S) & YEAR	TITLE	DESCRIPTION	LIMITATIONS
IEEE ACCESS (2024)	IoT-Based Smart Air Quality Monitoring System Using ESP8266	Developed a real-time monitoring system using ESP8266 and gas sensors	Limited accuracy due to low-cost sensors
ELSEVIER (2024)	Real-Time Air Quality Monitoring Using Low-Cost Sensors and IoT	Focuses on affordable sensors for continuous air monitoring	Calibration issues and environmental impact on readings
SPRINGER (2024)	Air Pollution Monitoring System Using IoT and Cloud	Uses cloud for data storage and visualization	Requires stable internet and increases complexity
IEEE SENSORS JOURNAL (2025)	Wireless Sensor Network-Based Air Quality Monitoring	Uses multiple sensors and wireless communication	High power consumption and maintenance cost
ELSEVIER (2024)	Indoor Air Quality Monitoring Using IoT	Monitors indoor pollution levels using IoT devices	Limited to indoor environments only
IEEE IOT JOURNAL (2025)	IoT and AI-Based Air Quality Prediction System	Combines IoT with AI for prediction of pollution trends	Complex implementation and higher cost
SPRINGER (2025)	Smart Environmental Monitoring Using IoT	Integrates multiple environmental parameters	Requires large data handling and processing

IJITEE (2024)	IoT-Based Air Quality Monitoring Using Arduino	Simple system using Arduino and sensors	No remote access and limited scalability
ELSEVIER (2024)	Multi-Gas Monitoring System Using IoT	Detects multiple gases for environmental safety	Sensor cross-sensitivity affects accuracy

Most of the studies reviewed focus on the use of microcontrollers such as Arduino and ESP8266 for developing air quality monitoring systems. These devices are widely preferred due to their low cost, ease of programming, and ability to interface with various sensors. The MQ-series gas sensors, particularly MQ-135, are commonly used for detecting harmful gases in the environment. These sensors enable continuous monitoring of air quality and provide data that can be processed and analyzed. The use of ESP8266 further enhances the system by enabling wireless communication through iFi, allowing data to be accessed remotely via web interfaces or mobile applications.

Another important observation from the literature is the increasing use of cloud platforms for data storage and analysis. Many researchers have integrated cloud computing with IoT systems to enable long-term monitoring and visualization of air quality data. Cloud-based systems allow users to access data from anywhere and provide features such as data logging, trend analysis, and reporting. However, these systems often require stable internet connectivity and may involve additional costs and complexity. This indicates a trade-off between advanced functionality and system simplicity.

The literature also highlights the use of wireless sensor networks (WSN) for large-scale air quality monitoring. These systems involve multiple sensor nodes deployed across different locations to collect environmental data. While WSN-based systems provide better coverage and improved accuracy, they require higher power consumption and maintenance. This makes them less suitable for small-scale or low-cost applications. On the other hand, simpler IoT-based systems using a single microcontroller and sensor are more practical for personal and localized monitoring.

Recent advancements in the field include the integration of artificial intelligence (AI) and machine learning (ML) techniques with IoT systems. These technologies are used to analyze collected data and predict future air quality trends. Predictive systems can help in taking preventive measures before pollution levels become hazardous. However, the implementation of AI-based systems increases the complexity and cost, making them less accessible for basic applications. This shows that while advanced technologies offer additional benefits, they may not always be necessary for simple monitoring purposes.

Another key finding from the literature is the importance of real-time monitoring and alert mechanisms. Many systems include features such as alarms, notifications, or buzzers to warn users when pollution levels exceed safe limits. This enhances user safety and ensures timely action. The use of web-based interfaces and mobile applications also improves user accessibility and interaction with the system. These features make IoT-based monitoring systems more practical and user-friendly compared to traditional methods.

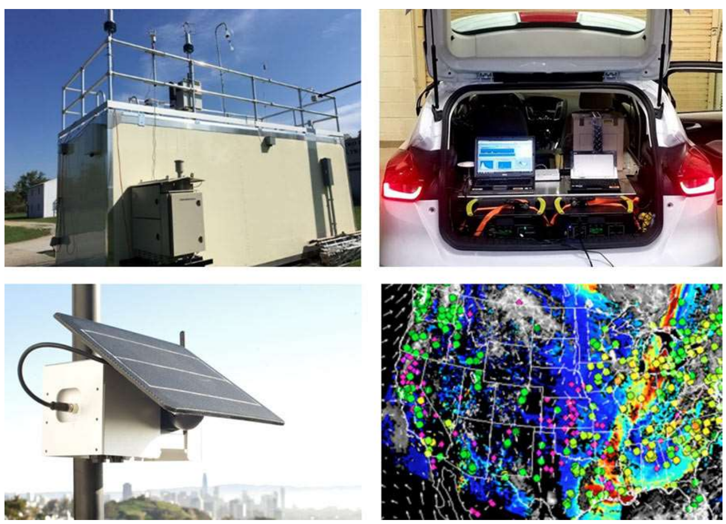
Despite the advantages, several limitations are identified in the reviewed studies. One of the major challenges is the accuracy of low-cost sensors, which may be affected by environmental factors such as temperature and humidity. Proper calibration is required to ensure reliable readings. Additionally, many systems depend on internet connectivity, which may not be available in all areas. Security and privacy concerns related to data transmission are also important issues that need to be addressed in IoT applications.

Overall, the literature review indicates that IoT-based air quality monitoring systems provide an effective and affordable solution for monitoring environmental conditions. These systems offer realtime data, remote accessibility, and ease of use, making them suitable for a wide range of applications. However, there is still a need to improve sensor accuracy, reduce system complexity, and enhance reliability. The reviewed studies provide a strong foundation for the development of improved systems that can address these challenges.

CHAPTER III
SYSTEM ANALYTICS AND REQUIREMENTS

3.1 EXSISTING SYSTEM

Air quality monitoring has been an essential practice for assessing environmental pollution and ensuring public health safety. Traditionally, air quality monitoring systems have been implemented using large-scale monitoring stations operated by government agencies and environmental organizations. These systems are equipped with advanced instruments capable of measuring various air pollutants such as carbon monoxide (CO), sulfur dioxide (SO_2), nitrogen oxides (NOx), ozone (O_3), and particulate matter (PM). The data collected from these stations is highly accurate and reliable, making them suitable for scientific analysis and policy-making. However, these systems are expensive to install and maintain, requiring significant infrastructure, skilled personnel, and regular calibration. Due to their high cost, they are usually installed only in selected locations such as major cities and industrial areas, limiting their accessibility to the general public.



One of the major limitations of traditional air quality monitoring systems is their inability to provide localized and real-time data for individual users. Since these monitoring stations are placed at fixed locations, they provide generalized data for a wide area rather than specific information for a particular household or small region. As a result, individuals are often unaware of the exact air quality in their immediate surroundings. This lack of localized monitoring reduces the effectiveness of these systems in helping people take timely preventive measures. Additionally, the data collected by these stations is often updated periodically rather than continuously, which may not reflect sudden changes in air quality caused by factors such as traffic congestion or industrial emissions.

With the advancement of embedded systems and sensor technologies, several low-cost air quality monitoring systems have been developed using microcontrollers such as Arduino.

These systems use gas sensors, particularly MQ-series sensors, to detect the presence of harmful gases in the environment. The sensor data is processed by the microcontroller and displayed on output devices such as LCD screens or serial monitors. These systems are more affordable compared to traditional monitoring stations and are suitable for small-scale applications. However, most of these systems lack connectivity features, meaning that the data can only be viewed locally. This limits their usefulness in modern applications where remote monitoring and data sharing are essential.

Another type of existing system is the GSM-based air quality monitoring system. In this approach, a GSM module is used to send air quality data to users via SMS alerts. This allows users to receive notifications about pollution levels without being physically present near the device. While this method provides a form of remote communication, it has several drawbacks. GSM modules increase the overall cost of the system and require a SIM card and network connectivity. Additionally, SMS-based communication is relatively slow and does not support continuous real-time monitoring. The information provided is limited and does not offer a detailed visualization of air quality data.

In recent years, cloud-based air quality monitoring systems have gained popularity. These systems use IoT technology to collect data from sensors and upload it to cloud platforms for storage and analysis. Users can access the data through web dashboards or mobile applications, allowing them to monitor air quality remotely. Cloud-based systems provide advanced features such as data logging, trend analysis, and visualization. They are particularly useful for large-scale monitoring and research purposes. However, these systems require a stable internet connection and may involve subscription costs for cloud services. The setup and maintenance of such systems can also be complex, making them less suitable for beginners or small-scale users.

Mobile application-based monitoring is another approach used in existing systems. These applications provide air quality information to users through smartphones. The data is usually obtained from centralized monitoring stations or external databases. While mobile apps are convenient and easy to use, they do not provide real-time data for specific locations. The information displayed is often generalized for a city or region, which may not accurately represent the air quality in a user's immediate environment. This limitation reduces the effectiveness of mobile apps in providing precise and actionable information.

Wireless sensor networks (WSN) have also been used for air quality monitoring in large-scale environments. In such systems, multiple sensor nodes are deployed across different locations to collect environmental data. The data is transmitted wirelessly to a central server for processing and analysis. WSN-based systems provide better coverage and more accurate data compared to single-node systems. However, they are complex to design and implement, requiring careful planning of network topology and communication protocols. They also consume more power and require regular maintenance, which increases operational costs.

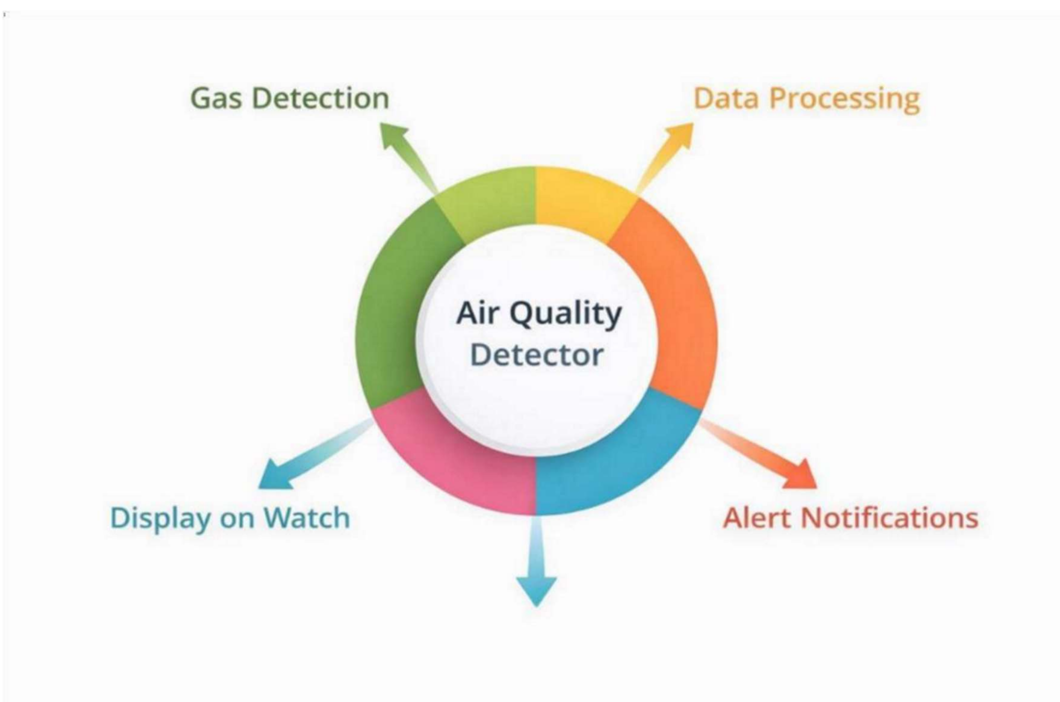
Despite the availability of these existing systems, several challenges remain. Many systems are either too expensive, complex, or not user-friendly. Some lack real-time monitoring capabilities, while others do not provide accurate localized data. Connectivity and accessibility are also major concerns, as many systems depend on internet or network availability. Additionally, the accuracy of low-cost sensors used in some systems can be affected by environmental factors such as temperature and humidity.

These limitations highlight the need for an improved air quality monitoring system that is cost-effective, portable, and capable of providing real-time data. The system should be easy to use, require minimal maintenance, and offer remote accessibility without significant complexity. This creates an opportunity for the development of IoT-based solutions that combine sensing technology, wireless communication, and user-friendly interfaces.

3.2 PROPOSED SYSTEM

The proposed system aims to design and develop a real-time air quality monitoring device using IoT technology to overcome the limitations of existing systems. The system utilizes the ESP8266 microcontroller and MQ-135 gas sensor to detect and monitor the level of air pollution in the surrounding environment. The main objective of this system is to provide an affordable, portable, and user-friendly solution that can continuously monitor air quality and display the data in real time through a web-based interface. Unlike traditional monitoring systems, which are expensive and limited in coverage, the proposed system is compact and suitable for both indoor and outdoor applications

The core component of the proposed system is the ESP8266 microcontroller, which acts as the processing unit and communication module. It is equipped with built-in WiFi capability, allowing it to connect to a local network and host a web server. The MQ-135 gas sensor is used to detect harmful gases such as carbon dioxide, ammonia, benzene, and smoke present in the air. The sensor continuously measures the concentration of these gases and sends the



data to the ESP8266 in the form of analog signals. The microcontroller processes this data and converts it into meaningful air quality values that can be easily understood by users.

One of the key features of the proposed system is real-time data monitoring and visualization. The ESP8266 hosts a web server and generates an IP address, which can be accessed through any web browser on devices such as smartphones, laptops, or tablets. When users enter this IP address, a web page is displayed showing the current air quality readings. This eliminates the need for additional software or mobile applications, making the system simple and accessible. The real-time updates ensure that users are always informed about the air quality in their surroundings, enabling them to take immediate action if necessary.

Another important feature of the proposed system is the implementation of an alert mechanism using a buzzer. The system is programmed with a predefined threshold value for air quality. When the detected pollution level exceeds this threshold, the buzzer is activated to alert the user. This feature provides an immediate warning about hazardous environmental conditions, enhancing user safety. The alert system is particularly useful in situations where users may not be actively monitoring the web interface, ensuring that they are still informed about critical changes in air quality.

The proposed system is designed to be cost-effective and easy to implement. By using low-cost components such as ESP8266, MQ-135 sensor, and basic electronic elements, the system can be developed at a minimal cost compared to traditional monitoring systems. This makes it accessible to a wide range of users, including students, researchers, and small organizations. The simplicity of the design also reduces the need for complex installation and maintenance, making it suitable for non-technical users.

Portability is another significant advantage of the proposed system. The compact size and lightweight components allow the device to be easily installed and moved to different locations as needed. This flexibility makes it suitable for various applications, including homes, offices, schools, and industrial environments. Users can monitor air quality in different areas without the need for multiple fixed monitoring stations.

The proposed system also emphasizes user-friendly operation. The web interface is designed to be simple and easy to understand, displaying air quality data in a clear and readable format. Users do not require any technical expertise to access or interpret the data. The system provides a seamless experience, from data collection to visualization, ensuring that users can effectively monitor their environment without difficulty.

In addition to its current features, the proposed system has the potential for future enhancements. It can be extended by integrating additional sensors to measure other environmental parameters such as temperature, humidity, and particulate matter. The system can also be connected to cloud platforms for data storage and analysis, enabling long-term monitoring and trend analysis. Mobile applications can be developed to provide notifications and improve user interaction. These enhancements can further improve the functionality and scalability of the system.

Despite its advantages, the proposed system has certain limitations. The accuracy of the MQ-135 sensor may vary depending on environmental conditions and requires proper calibration for reliable performance. The system also depends on the availability of a stable WiFi connection for data transmission. However, these limitations can be addressed through proper design and future improvements.

3.3 ADVANTAGES OF PROPOSED SYSTEM

1. Real-Time Monitoring Capability

One of the major advantages of the proposed air quality monitoring system is its ability to provide real-time data. The MQ-135 sensor continuously detects the concentration of harmful gases present in the environment and sends the data to the ESP8266 microcontroller for processing. This enables the system to update air quality readings instantly without any delay. Real-time monitoring is highly important in environments where pollution levels can change rapidly, such as industrial areas, traffic zones, and enclosed indoor spaces. By providing immediate updates, the system allows users to take quick preventive measures, thereby reducing health risks. Unlike traditional monitoring systems that provide periodic updates, this system ensures continuous observation of environmental conditions.

2. Cost-Effective Solution

The proposed system is designed to be affordable and accessible to a wide range of users. It utilizes low-cost components such as the ESP8266 microcontroller, MQ-135 gas sensor, and basic electronic devices like a buzzer. Compared to traditional air quality monitoring stations, which require expensive equipment and infrastructure, this system can be developed at a significantly lower cost. This makes it suitable for students, small organizations, and individuals who require a simple monitoring solution. The cost-effectiveness of the system does not compromise its functionality, as it still provides essential features such as real-time monitoring and alert mechanisms.

3. Wireless Connectivity and Remote Access

Another key advantage of the proposed system is its wireless communication capability. The ESP8266 microcontroller comes with built-in WiFi functionality, which allows the system to connect to a local network and host a web server. Users can access the air quality data by entering the device's IP address in a web browser. This enables remote monitoring without the need for additional software or hardware. The ability to access data from smartphones, laptops, or tablets makes the system highly convenient and user-friendly. This feature is particularly useful in modern IoT environments where remote access and data sharing are essential.

4. User-Friendly Interface

The proposed system is designed with simplicity and ease of use in mind. The web-based interface provides a clear and straightforward display of air quality data, making it easy for users to understand the information. Unlike complex monitoring systems that require technical expertise, this system can be used by individuals with minimal knowledge of technology. The interface eliminates the need for specialized applications, as it can be accessed through any standard web browser. This user-friendly design ensures that the system is accessible to a broader audience, including non-technical users.

5. Immediate Alert System

The integration of a buzzer in the system provides an effective alert mechanism. When the air quality exceeds a predefined threshold level, the buzzer is activated to notify the user immediately. This feature enhances the safety of the system by ensuring that users are alerted to hazardous conditions even if they are not actively monitoring the data on the web page. The alert system is particularly useful in critical situations where quick response is required to prevent health risks. It adds an extra layer of functionality to the system, making it more reliable and practical.

6. Portability and Flexibility

The compact design of the proposed system makes it highly portable and flexible. The device can be easily installed in different environments such as homes, offices, schools, and industrial areas. Unlike traditional monitoring systems that are fixed in specific locations, this system can be moved and used in multiple places as needed. This flexibility allows users to monitor air quality in various settings without requiring multiple devices. The portability of the system also makes it suitable for temporary installations and field applications.

7. Easy Installation and Maintenance

The proposed system is simple to set up and does not require complex installation procedures. The components can be easily connected using basic circuit design techniques, and the system can be programmed using Arduino IDE. Maintenance requirements are minimal, as the system uses readily available components that can be easily replaced if needed. This reduces the overall effort and cost associated with system operation. The simplicity of installation and maintenance makes the system suitable for beginners and educational purposes.

8. Energy Efficient Operation

The system is designed to operate efficiently with low power consumption. The ESP8266 microcontroller and MQ-135 sensor require minimal power, making the system suitable for continuous operation. This energy efficiency is beneficial for long-term monitoring applications and can also support battery-powered operation if needed. Low power consumption reduces operational costs and enhances the sustainability of the system.

9. Scalability and Future Enhancement

The proposed system is flexible and can be easily expanded to include additional features. More sensors can be integrated to measure other environmental parameters such as temperature, humidity, and particulate matter. The system can also be connected to cloud platforms for data storage and analysis. Mobile applications and advanced data processing techniques can be added to enhance functionality. This scalability ensures that the system can evolve with future technological advancements and meet changing user requirements.

10. Promotes Environmental Awareness

One of the significant advantages of the proposed system is its contribution to environmental awareness. By providing real-time information about air quality, the system helps users understand the level of pollution in their surroundings. This encourages individuals to take necessary actions to reduce exposure to harmful pollutants and adopt environmentally friendly practices. The system plays an important role in educating users about the importance of maintaining a clean and healthy environment.

3.4 SYSTEM REQUIREMENTS

HARDWARE REQIUREMENTS

Microcontroller:ESP8266 (NodeMCU)

Sensor Module: MQ-135 Gas Sensor

Alert System: Buzzer

Display Unit: Web-based display (via IP address in browser)

Power Supply: 5V DC (USB cable or external supply)

Connectivity: Built-in Wi-Fi (ESP8266)

SOFTWARE REQIUREMENTS

Programming Platform: ArduinoIDE

Programming Language: Embedded C/C++

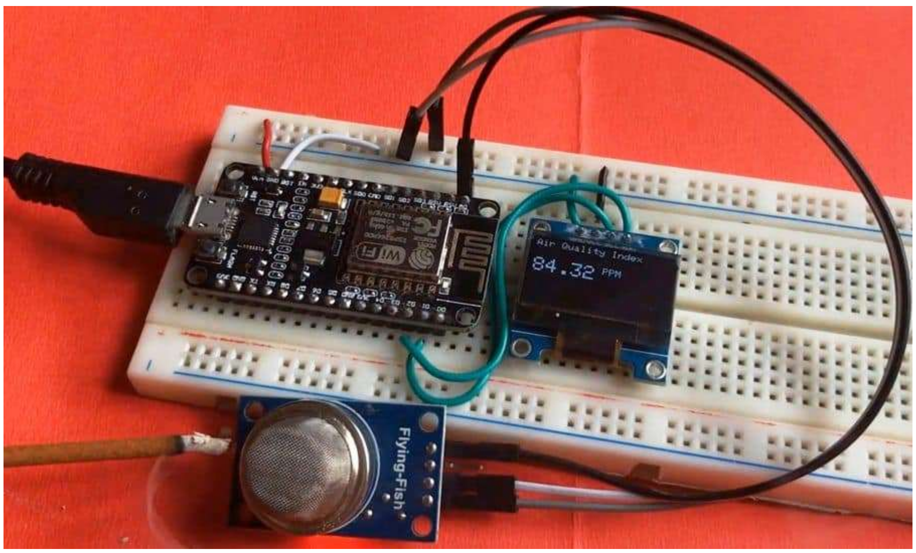
Web Technology: HTML (for webpage display)

Communication Protocol: HTTP (for web server communication)

Library Support: ESP8266 WiFi Library

Browser: Google Chrome / Any Web Browser

Operating System: Windows / Linux



CHAPTER IV

SYSTEM DESIGN AND IMPLEMENTATION

4.1 DEIGN

1. System Design Overview

The design of the proposed air quality monitoring system focuses on creating a simple, efficient, and cost-effective solution using IoT technology. The system is developed using the ESP8266 microcontroller, MQ-135 gas sensor, and a buzzer for alert purposes. The overall design integrates sensing, processing, communication, and alert mechanisms into a single compact system. The MQ-135 sensor is responsible for detecting harmful gases present in the environment, while the ESP8266 processes the sensor data and transmits it through WiFi. The system is designed to host a web server, allowing users to access air quality data through a web browser using the device's IP address. The design ensures real-time monitoring, easy accessibility, and user-friendly operation.

2. Hardware Design

The hardware design of the system involves the proper selection and interconnection of components. The MQ-135 gas sensor is connected to the ESP8266 microcontroller to provide analog input representing the air quality level. The sensor continuously monitors the environment and sends data to the microcontroller. A buzzer is also connected to the ESP8266, which acts as an alert system when pollution levels exceed a predefined threshold. The connections are made using a breadboard and jumper wires, allowing easy assembly and modification of the circuit. The system is powered using a 5V DC supply through a USB cable or external power source. The hardware design ensures minimal complexity while maintaining efficient performance.

3. Software Design

The software design plays a crucial role in the functioning of the system. The ESP8266 is programmed using the Arduino IDE with Embedded C/C++ language. The program includes functions for reading sensor data, processing values, and controlling the buzzer. It also includes WiFi configuration code to connect the ESP8266 to a local network. Once connected, the microcontroller hosts a web server that displays air quality data on a web page. The HTML code for the web page is embedded within the program, allowing the ESP8266 to serve the page directly to the user's browser. The software design ensures smooth communication between hardware components and provides real-time updates to the user.

4. Communication Design

The communication design of the system is based on WiFi technology. The ESP8266 connects to a local WiFi network and acts as a web server. It generates an IP address, which users can enter into a web browser to access the air quality data. The communication between the ESP8266 and the user's device is carried out using the HTTP protocol. This enables seamless data transfer and real-time monitoring without the need for additional software or applications. The communication design ensures that the system is accessible from multiple devices within the network, enhancing its usability.

5. User Interface Design

The user interface of the system is designed to be simple and easy to understand. A web-based interface is used to display the air quality data. The interface shows real-time readings in a clear format, allowing users to quickly interpret the information. The design avoids complexity and focuses on providing essential data in a readable manner. Since the interface is accessed through a web browser, users do not need to install any additional software. This improves accessibility and ensures that the system can be used by individuals with minimal technical knowledge.

6. **Alert System Design**

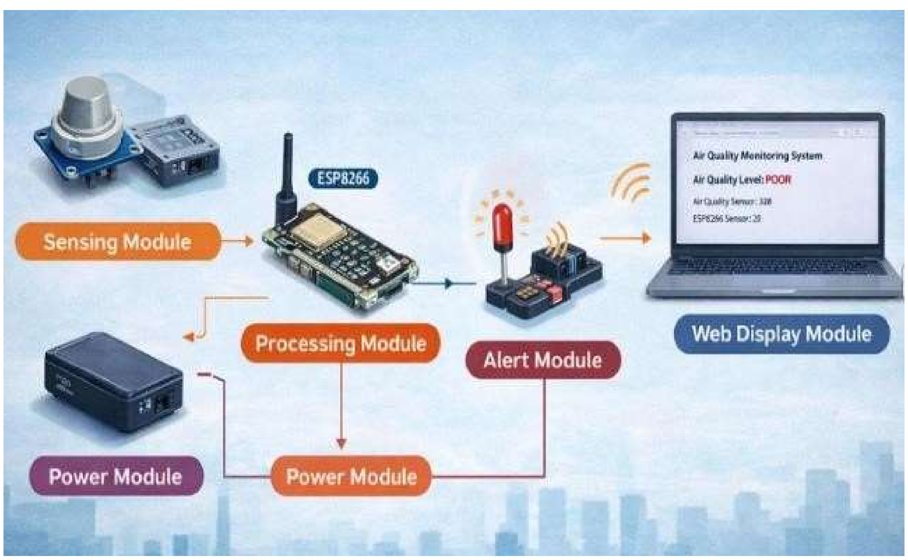
The alert system is an important part of the overall design. A buzzer is used to provide an audible warning when the air quality exceeds a predefined threshold. The ESP8266 continuously compares the sensor readings with the threshold value. If the pollution level crosses the limit, the buzzer is activated automatically. This feature ensures that users are alerted to dangerous conditions even if they are not actively monitoring the web page. The alert system enhances the safety and reliability of the system.

7. **Power Management Design**

The power management design ensures that the system operates efficiently with a stable power supply. The ESP8266 and MQ-135 sensor require a consistent voltage to function properly. The system is powered using a 5V

DC supply, which can be provided through a USB cable or external adapter. Proper power management helps in maintaining the performance and reliability of the system. It also ensures that the system can operate continuously without interruptions.

4.2 SYSTEM ARCHITECTURE DIAGRAM



The system architecture of the proposed IoT-based air quality monitoring device represents the overall structure and interaction between different components used in the system. It provides a clear understanding of how data flows from the sensing unit to the end user through processing and communication modules. The architecture is designed to be simple, efficient, and scalable, ensuring that all components work together seamlessly to provide real-time air quality monitoring. The main elements of the architecture include the MQ-135 gas sensor, ESP8266 microcontroller, buzzer alert system, WiFi communication module, and web-based user interface.

At the initial stage of the system architecture, the MQ-135 gas sensor acts as the sensing unit. It continuously monitors the surrounding air and detects the presence of harmful gases such as carbon dioxide, ammonia, and smoke. The sensor operates by sensing changes in air composition and converting them into electrical signals. These signals are in analog form and represent the concentration of pollutants in the environment. The accuracy of the sensor depends on proper calibration and environmental conditions. The continuous sensing process ensures that real-time data is always available for further processing.

The next stage in the architecture is the data processing unit, which is handled by the ESP8266 microcontroller. The analog signals from the MQ-135 sensor are sent to the microcontroller, where they are converted into digital values. The ESP8266 processes these values using programmed algorithms to determine the air quality level. It compares the sensor readings with predefined threshold values to identify whether the air quality is safe or hazardous. The microcontroller acts as the brain of the system, controlling all operations and coordinating between different modules.

Once the data is processed, the system moves to the communication stage. The ESP8266 uses its built-in WiFi capability to connect to a local network. It acts as a web server and generates an IP address, which serves as the access point for users. The processed air quality data is transmitted over the network using the HTTP protocol. This allows users to access the data through a web browser on devices such as smartphones, laptops, or tablets. The communication module ensures fast and efficient data transfer, enabling real-time monitoring without delays.

The system architecture also includes a user interface component in the form of a web-based display. The ESP8266 hosts a web page that presents the air quality data in a simple and readable format. When users enter the IP address into their browser, the web page is displayed with real-time updates. This eliminates the need for additional software or mobile applications. The web interface is designed to be user-friendly, ensuring that users can easily understand the information provided by the system.

An important part of the system architecture is the alert mechanism. A buzzer is integrated into the system to provide immediate notification when the air quality exceeds safe limits. The ESP8266 continuously monitors the processed data and compares it with a predefined threshold. If the pollution level crosses this threshold, the microcontroller activates the buzzer. This alert system ensures that users are informed about hazardous conditions even if they are not actively monitoring the web interface. It enhances the safety and reliability of the system.

The power supply unit is another essential component of the system architecture. The ESP8266 and MQ-135 sensor require a stable power source to function properly. The system is powered using a 5V DC supply, which can be provided through a USB connection or an external adapter. Proper power management ensures consistent performance and prevents system failures due to voltage fluctuations. The power supply unit supports continuous operation, enabling the system to monitor air quality without interruptions.

The overall architecture follows a structured data flow, starting from data acquisition, followed by data processing, communication, and finally data visualization. Each component in the system has a specific role, and their integration ensures smooth operation. The modular design allows for easy modification and expansion of the system. Additional sensors or features can be incorporated in the future without significantly altering the existing architecture.

One of the key advantages of this system architecture is its simplicity and efficiency. Unlike complex traditional systems, the proposed architecture uses minimal components while still providing essential functionalities such as real-time monitoring, remote access, and alert mechanisms. The use of ESP8266 eliminates the need for separate communication modules, reducing both cost and complexity. The architecture is also scalable, allowing future enhancements such as cloud integration, mobile applications, and advanced data analysis.

In conclusion, the system architecture of the proposed air quality monitoring device provides a clear and organized framework for the operation of the system. It effectively integrates sensing, processing, communication, and alert mechanisms to deliver real-time air quality data to users. The design is simple, cost-effective, and user-friendly, making it suitable for a wide range of applications. The architecture not only meets the current requirements but also provides flexibility for future improvements, ensuring long-term usability and relevance in the field of IoT-based environmental monitoring.

4.3 MODULE DESIGN

1. Sensor Module Design

The sensor module is designed to detect and measure the concentration of harmful gases present in the surrounding environment. In this system, the MQ-135 gas sensor is used due to its ability to sense gases such as carbon dioxide, ammonia, benzene, and smoke. The sensor operates based on changes in resistance when exposed to different gas concentrations. These variations are converted into analog voltage signals, which represent the air quality level. The sensor is connected to the ESP8266 microcontroller, and its output is continuously monitored. Proper calibration is an important aspect of the sensor module design to ensure accurate readings. Environmental conditions such as temperature and humidity can influence sensor performance, and these factors are considered during the design process.

The sensor module is placed in a position where it can effectively capture air samples, ensuring reliable and real-time monitoring.

2. Processing Module Design

The processing module is responsible for handling the data received from the sensor and making decisions based on predefined conditions. The ESP8266 microcontroller serves as the core processing unit in this design. It reads the analog input from the MQ-135 sensor and converts it into digital values using its internal processing capabilities. The microcontroller is programmed using Arduino IDE, where the logic for data processing, threshold comparison, and control operations is implemented. The processed data is used to determine the air quality status, such as safe or unsafe levels. The processing module also controls the operation of other modules, including the buzzer and web server. Efficient programming ensures that the system responds quickly to changes in air quality and maintains accurate data processing.

3. Communication Module Design

The communication module is designed to enable data transmission between the system and the user. The ESP8266 microcontroller includes built-in WiFi functionality, which eliminates the need for an external communication module. In this design, the ESP8266 connects to a local WiFi network and acts as a web server. It generates an IP address, which users can use to access the system through a web browser. The communication between the ESP8266 and the user's device is carried out using the HTTP protocol. This allows real-time data transfer and remote monitoring of air quality. The communication module is designed to be efficient and reliable, ensuring that data is transmitted without significant delays or losses.

4. Alert Module Design

The alert module is designed to provide immediate notification when the air quality exceeds safe limits. A buzzer is used as the alert device in this system. The ESP8266 continuously compares the sensor readings with a predefined threshold value. When the pollution level crosses this threshold, the microcontroller activates the buzzer to produce an audible alert. This ensures that users are warned about hazardous conditions even if they are not actively monitoring the web interface. The alert module is simple yet effective, enhancing the safety and responsiveness of the system. The threshold values can be adjusted based on user requirements or environmental standards.

5. Display Module Design

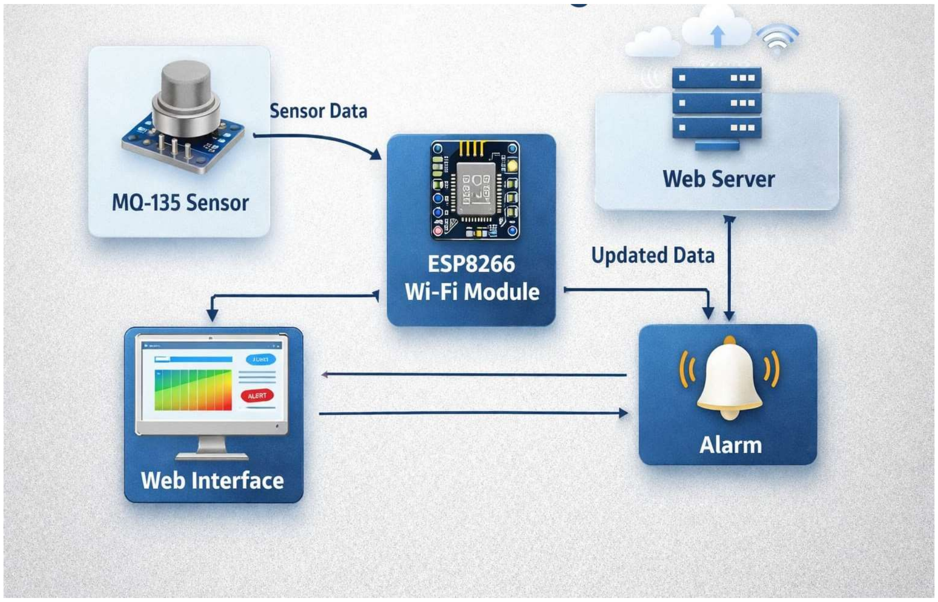
The display module is designed to present the air quality data to the user in an easy-to-understand format. Instead of using a physical display like an LCD, this system uses a web-based interface. The ESP8266 hosts a web page that displays real-time air quality readings. Users can access this web page by entering the IP address of the device into a web browser. The interface is designed using basic HTML, ensuring simplicity and compatibility with different devices. The display module provides clear and readable information, allowing users to quickly understand the air quality status. This approach reduces hardware complexity and cost while improving accessibility.

6.Power Supply Module Design

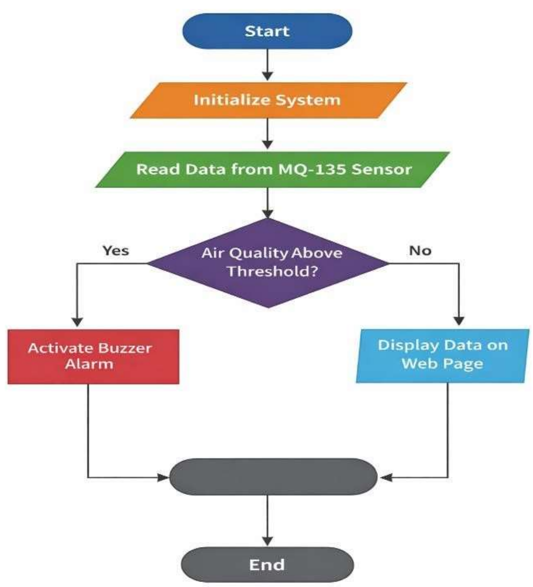
The power supply module is designed to provide a stable and reliable source of energy for the system. The ESP8266 and MQ-135 sensor require a regulated 5V DC supply for proper operation. The system can be powered using a USB cable connected to a computer or an external adapter. Proper voltage regulation is essential to prevent damage to the components and ensure consistent performance. The power supply module is designed to support continuous operation, enabling the system to function without interruptions. Efficient power management also contributes to the overall reliability of the system.

7.Integration of Modules

The integration of all modules is a critical aspect of the system design. Each module is interconnected to ensure smooth data flow and operation. The sensor module collects data and sends it to the processing module, which analyzes the information and makes decisions. The communication module transmits the processed data to the user, while the display module presents it in a readable format. The alert module provides notifications in case of hazardous conditions, and the power supply module ensures continuous operation. The integration is designed to be seamless, allowing all components to work together efficiently. This modular approach also makes it easier to troubleshoot, maintain, and upgrade the system.



4.4 FLOW CHART



The flow of the IoT-based air quality monitoring system begins with the initialization stage, where the ESP8266 microcontroller is powered on and configured. During this stage, the system initializes all components including the MQ-135 gas sensor, buzzer, and WiFi connection. The ESP8266 connects to the local WiFi network and generates an IP address, which will later be used to access the web interface. Once the initialization is complete, the system enters into a continuous monitoring loop where it starts reading data from the gas sensor.

In the next stage, the MQ-135 sensor continuously senses the air quality and sends analog data to the ESP8266. The microcontroller processes this data and converts it into meaningful values representing pollution levels. These values are then compared with a predefined threshold to determine whether the air quality is safe or unsafe. If the detected gas concentration exceeds the threshold level, the system identifies it as a hazardous condition. Otherwise, it continues monitoring without triggering any alert. This decision-making step is a critical part of the system as it ensures timely detection of harmful environmental conditions.

Finally, the processed data is displayed on a web page hosted by the ESP8266, which can be accessed through a browser using the generated IP address. At the same time, if the air quality is found to be unsafe, the buzzer is activated to alert the user immediately. The system continues to repeat this cycle, ensuring real-time monitoring and continuous updates. This loop-based operation makes the system efficient, responsive, and capable of providing instant feedback to users about the air quality in their surroundings.

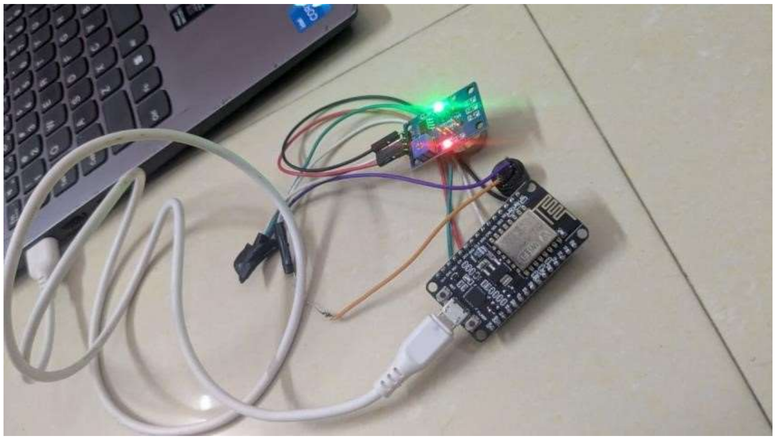
4.5 SYSTEM IMPLEMENTATION

1. Overview of Implementation

The system implementation of the IoT-based air quality monitoring device involves the practical realization of the proposed design using hardware and software components. The implementation phase focuses on integrating the MQ-135 gas sensor, ESP8266 microcontroller, buzzer, and web interface to create a fully functional system. The main objective of this phase is to ensure that all components work together effectively to monitor air quality in real time and provide accurate data to the user. The implementation process includes hardware setup, software programming, system integration, and testing. Each step is carefully executed to achieve a reliable and efficient monitoring system.

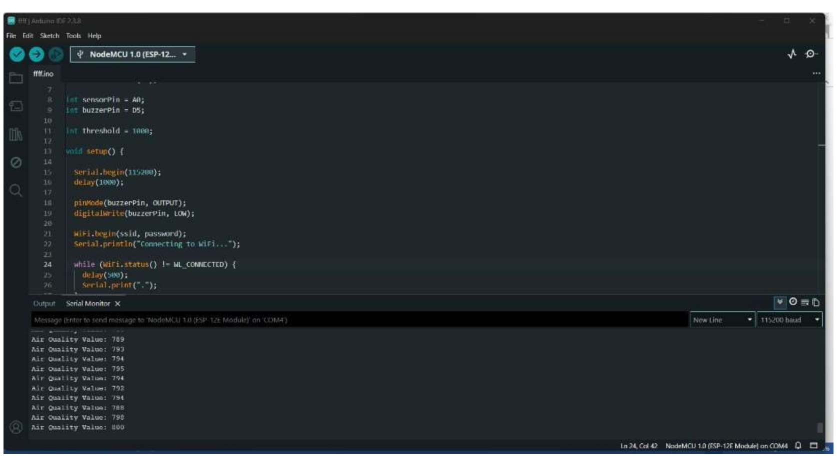
2. Hardware Implementation

The hardware implementation involves assembling and connecting all the physical components required for the system. The MQ-135 gas sensor is connected to the ESP8266 microcontroller to measure air quality levels. The sensor's output pin is interfaced with the microcontroller to provide continuous data readings. A buzzer is also connected to the ESP8266 to act as an alert system when pollution levels exceed the threshold. The connections are made using a breadboard and jumper wires, allowing easy setup and modifications. The entire system is powered using a 5V DC supply, which can be provided through a USB cable or an external adapter. Proper connections and stable power supply are essential to ensure accurate sensor readings and reliable system performance.



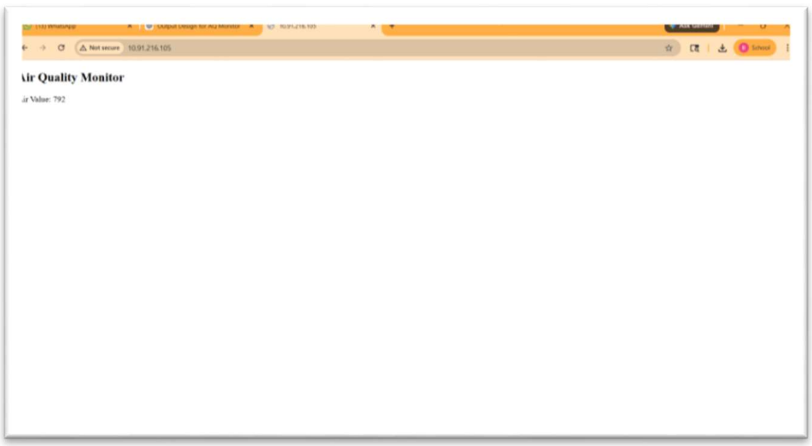
3.Software Implementation

The software implementation is carried out using the Arduino IDE, where the ESP8266 microcontroller is programmed using Embedded C/C++ language. The program includes functions for initializing the system, reading sensor data, processing values, and controlling the buzzer. It also includes WiFi configuration code that connects the ESP8266 to a local network. Once connected, the microcontroller acts as a web server and hosts a web page displaying air quality data. The HTML code for the web interface is embedded within the program, enabling the ESP8266 to serve the page directly to the user's browser. The software ensures continuous data acquisition and realtime updates.



4.WiFi Configuration and Web Server Setup

An important part of the system implementation is configuring the WiFi connection and setting up the web server. The ESP8266 is programmed with the SSID and password of the local WiFi network. Once connected, it obtains an IP address, which is displayed on the serial monitor. This IP address is used to access the web interface. The ESP8266 acts as a server and responds to client requests by sending the latest air quality data. The web page is designed using basic HTML to display sensor readings in a clear and user-friendly format. This approach eliminates the need for additional applications and allows easy access through any web browser.



5.Sensor Data Processing and Calibration

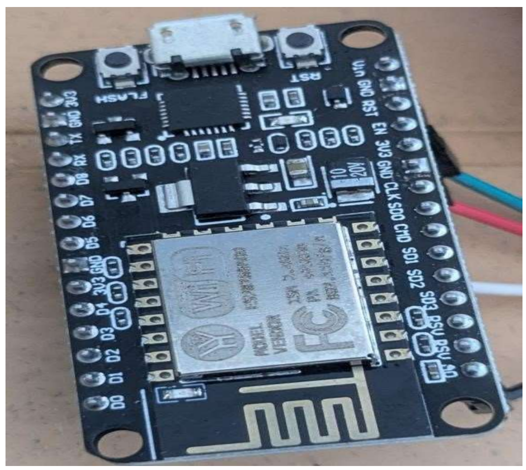
The sensor data processing is a critical step in the implementation. The MQ-135 sensor provides analog output, which is read by the ESP8266 and converted into digital values. These values are processed to determine the level of air pollution. Calibration is required to ensure that the sensor provides accurate readings. This involves adjusting the sensor values based on environmental conditions and reference standards. Proper calibration improves the reliability of the system and ensures that the data reflects actual air quality conditions.

6.Alert System Implementation

The alert system is implemented using a buzzer connected to the ESP8266 microcontroller. A threshold value for air quality is defined in the program. When the sensor readings exceed this threshold, the microcontroller activates the buzzer to alert the user. This feature ensures immediate notification of hazardous conditions. The alert system is simple yet effective, providing an additional layer of safety. It is particularly useful in situations where users may not be actively monitoring the web interface.

7.System Testing and Validation

After assembling and programming the system, testing is carried out to ensure proper functionality. The system is tested under different environmental conditions to verify its performance. The sensor readings are observed and compared with expected values to check accuracy. The WiFi connectivity and web interface are also tested to ensure smooth communication and data display. The buzzer is tested by simulating high pollution conditions to confirm that the alert system works correctly. Any errors or issues identified during testing are corrected to improve system performance.



8.Integration and Final Deployment

The final stage of implementation involves integrating all components into a single system and deploying it for practical use. The device is placed in the environment where air quality needs to be monitored. Once deployed, the system continuously collects data, processes it, and displays it on the web interface. Users can access the data through the IP address and monitor air quality in real time. The integrated system operates efficiently, providing reliable and continuous monitoring. The modular design allows easy maintenance and future upgrades.

4.6 WORKING OF THE SYSTEM

The working of the proposed IoT-based air quality monitoring system is based on continuous sensing, processing, and real-time data transmission using embedded and wireless technologies. The system operates in a cyclic manner, where each component performs a specific function to ensure accurate and efficient monitoring of air quality. The main components involved in the working process include the MQ-135 gas sensor, ESP8266 microcontroller, WiFi communication module, buzzer alert system, and web-based interface. The integration of these components enables the system to provide real-time air quality data and immediate alerts when pollution levels exceed safe limits.

The operation of the system begins with the initialization stage. When the device is powered on, the ESP8266 microcontroller initializes all connected components, including the MQ-135 sensor and the buzzer. At the same time, the ESP8266 attempts to connect to a predefined WiFi network using the stored SSID and password. Once the connection is successfully established, the microcontroller generates an IP address, which is displayed on the serial monitor. This IP address acts as the access point for users to view air quality data through a web browser. After completing the initialization process, the system enters a continuous monitoring loop.

In the next stage, the MQ-135 gas sensor starts sensing the surrounding air. The sensor detects various harmful gases such as carbon dioxide, ammonia, benzene, and smoke. It works by measuring changes in resistance when exposed to different gas concentrations. These changes are converted into analog voltage signals, which are then sent to the ESP8266 microcontroller. The sensor continuously collects data from the environment, ensuring that the system always has up-to-date information about air quality.

The ESP8266 microcontroller receives the analog signals from the sensor and processes them into meaningful values. This involves converting the raw sensor data into digital values and interpreting them to determine the level of air pollution. The processed data is then compared with predefined threshold values stored in the program. These threshold values represent safe and unsafe levels of air quality. If the sensor reading is below the threshold, the air quality is considered safe, and the system continues normal operation. If the reading exceeds the threshold, the system identifies it as a hazardous condition.

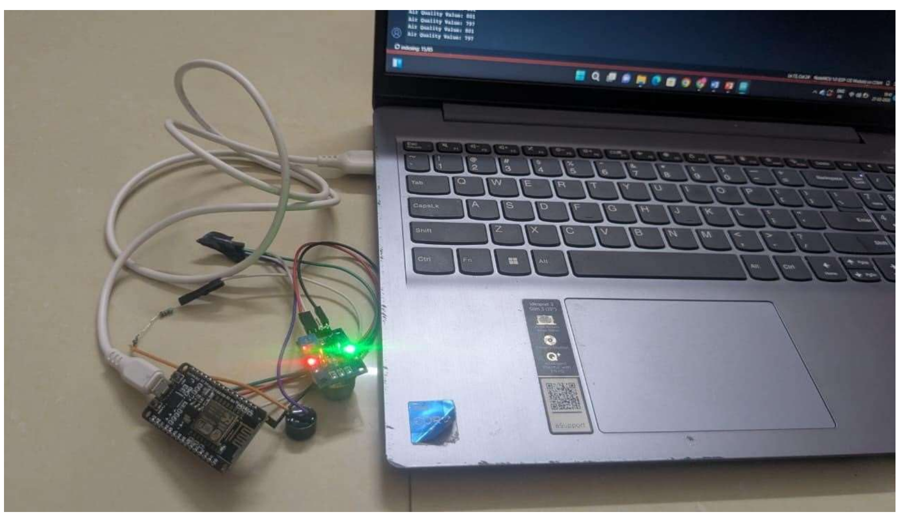
When a hazardous condition is detected, the alert mechanism is activated. The ESP8266 sends a signal to the buzzer, which produces an audible sound to notify the user. This ensures that users are immediately alerted to dangerous air quality levels, even if they are not actively monitoring the system. The buzzer continues to sound until the pollution level returns to a safe range or the system is reset. This feature enhances the safety of the system by providing realtime alerts.

Simultaneously, the processed air quality data is transmitted to the user through the web interface. The ESP8266 acts as a web server and hosts a web page that displays the air quality readings. Users can access this web page by entering the IP address into a web browser on their smartphone, laptop, or tablet. The web page provides real-time updates, allowing users to monitor air quality continuously. The interface is designed to be simple and easy to understand, ensuring that users can quickly interpret the data.

The system operates in a continuous loop, where sensing, processing, alerting, and data transmission occur repeatedly. This ensures that the system provides up-to-date information at all times. The loop-based operation allows the system to respond quickly to changes in air quality, making it highly efficient and reliable. The use of WiFi communication ensures that data is transmitted instantly, enabling real-time monitoring without delays.

Power management is also an important aspect of the system's working. The device is powered using a stable 5V DC supply, which ensures consistent performance of the components. The ESP8266 and sensor operate with low power consumption, allowing the system to run continuously for extended periods. Proper power supply ensures that the system does not experience interruptions or fluctuations during operation.

Overall, the working of the system demonstrates the effective integration of sensing, processing, and communication technologies. The MQ-135 sensor continuously monitors the environment, the ESP8266 processes the data and controls system operations, the buzzer provides alerts, and the web interface enables user interaction. The system successfully achieves real-time air quality monitoring with remote accessibility and immediate alert mechanisms.



CHAPTER V

RESULTS AND OUTPUT EVALUATION

5.1 EVALUATION METRICS EXPLANATION

1.Accuracy of Sensor Readings

Accuracy is one of the most important evaluation metrics for an air quality monitoring system. It refers to how closely the sensor readings match the actual environmental conditions. In this project, the MQ-135 gas sensor is used to detect harmful gases, and its performance is evaluated based on its ability to provide reliable data. Although the sensor is low-cost, it is capable of detecting variations in air quality effectively. The accuracy of the system depends on proper calibration and stable environmental conditions. During testing, the sensor readings showed consistent changes when exposed to different levels of pollution, indicating that the system can effectively identify variations in air quality. However, slight deviations may occur due to factors such as temperature and humidity. Overall, the system provides sufficient accuracy for general monitoring purposes.

2.Response Time

Response time refers to the speed at which the system detects changes in air quality and reflects those changes in the output. A faster response time is essential for real-time monitoring systems. In this project, the MQ-135 sensor continuously senses the environment, and the ESP8266 processes the data instantly. The system updates the web interface in real time, allowing users to observe changes without delay. The buzzer is also activated immediately when the pollution level exceeds the threshold. The quick response of the system ensures timely alerts, which is crucial for preventing exposure to harmful conditions. The evaluation shows that the system responds efficiently to sudden changes in air quality.

3.Reliability of the System

Reliability refers to the ability of the system to perform consistently over time without failure. The proposed system is designed to operate continuously, providing real-time air quality monitoring. During testing, the system was observed to function without interruptions for extended periods. The ESP8266 microcontroller efficiently handled data processing, communication, and alert operations simultaneously. The hardware components showed stable performance, and the system maintained consistent output. This demonstrates that the system is reliable and suitable for long-term monitoring applications.

4.Data Transmission Efficiency

Data transmission efficiency is an important metric for evaluating the performance of IoT-based systems. It refers to how effectively the system transmits data from the sensor to the user through the network. In this project, the ESP8266 uses WiFi to send data to a web interface. The system was tested for network performance, and it was observed that data was transmitted quickly and without significant delay. The web page displayed real-time updates, and multiple devices were able to access the data simultaneously. This indicates that the communication module is efficient and capable of handling real-time data transmission.

5.Alert System Performance

The performance of the alert system is evaluated based on its ability to notify users when air quality exceeds safe limits. The buzzer serves as the alert mechanism in this project. When the sensor detects high pollution levels, the ESP8266 activates the buzzer immediately. During testing, the alert system responded accurately to hazardous conditions and remained inactive during safe conditions. This ensures that users receive timely warnings without unnecessary disturbances. The alert system enhances the overall effectiveness of the monitoring system.

6.Power Consumption

Power consumption is an important factor, especially for systems designed for continuous operation. The ESP8266 microcontroller and MQ-135 sensor are known for their low power requirements. The system was evaluated for energy efficiency, and it was observed that it consumes minimal power while maintaining continuous operation. This makes the system suitable for long-term use and even for battery-powered applications. Efficient power consumption reduces operational costs and improves system sustainability.

7.User Interface Efficiency

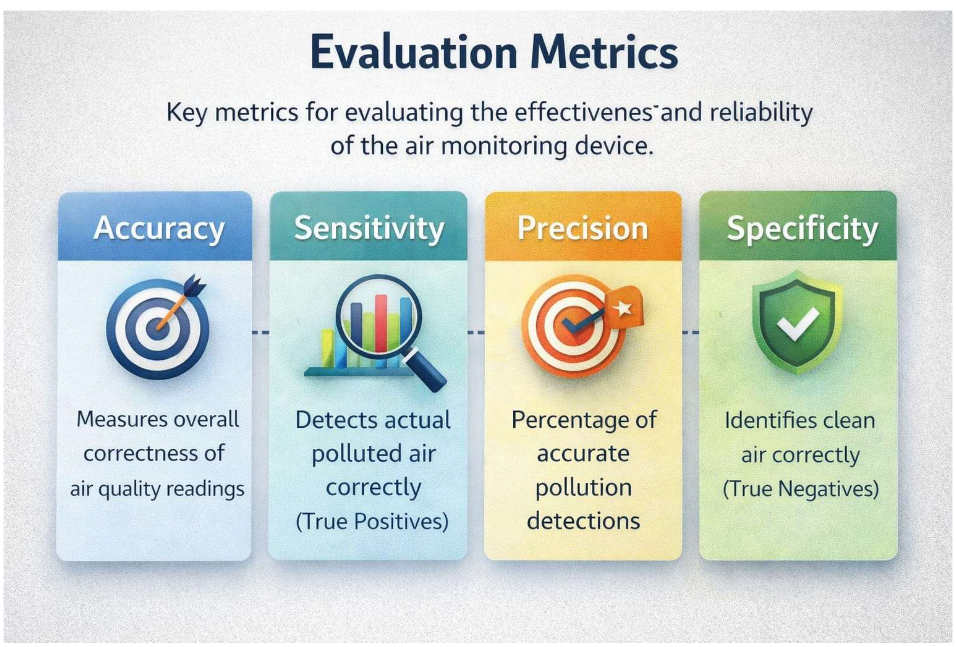
The efficiency of the user interface is evaluated based on its usability and accessibility. The proposed system uses a web-based interface to display air quality data. The interface is simple, clear, and easy to understand. Users can access the data through any web browser by entering the IP address of the device. The real-time updates and straightforward design make it easy for users to interpret the information. The absence of complex features ensures that even nontechnical users can use the system effectively. This enhances the overall user experience.

8.Scalability of the System

Scalability refers to the ability of the system to be expanded or upgraded to include additional features. The proposed system is designed with a modular architecture, allowing easy integration of additional sensors and functionalities. For example, sensors for temperature, humidity, and particulate matter can be added to enhance the system. The system can also be connected to cloud platforms for data storage and analysis. This flexibility makes the system scalable and adaptable to future requirements.

9.Cost Efficiency

Cost efficiency is an important metric for evaluating the practicality of the system. The proposed system is developed using low-cost components such as ESP8266, MQ-135 sensor, and basic electronic components. Compared to traditional air quality monitoring systems, the cost of this system is significantly lower. Despite its affordability, the system provides essential features such as real-time monitoring, remote access, and alert mechanisms. This makes it a cost-effective solution for a wide range of users.



5.2 ANALYSIS OF RESULT

The analysis of results is an important phase in evaluating the performance and effectiveness of the proposed IoT-based air quality monitoring system. This section focuses on interpreting the data obtained from the system and understanding how well it meets the intended objectives. The system was tested under different environmental conditions to observe its behavior, accuracy, and responsiveness. The results obtained from the MQ-135 gas sensor and displayed through the web interface provide valuable insights into the system's functionality and reliability.

The system demonstrated effective performance in detecting variations in air quality. When tested in a clean environment, such as a well-ventilated room, the sensor readings remained low and stable, indicating safe air conditions. This shows that the system can accurately identify normal environmental conditions without generating unnecessary alerts. The consistency in readings confirms that the sensor and microcontroller are functioning properly and that the system is capable of continuous monitoring without fluctuations.

In moderately polluted environments, such as areas with cooking activities or limited ventilation, the sensor readings showed a noticeable increase. The system successfully captured these changes and displayed them on the web interface in real time. The gradual increase in values indicated that the system is sensitive to changes in gas concentration. However, the readings remained below the predefined threshold, and the buzzer was not activated. This demonstrates that the system can differentiate between safe and moderately polluted conditions, avoiding false alarms and ensuring reliable performance.

During testing in highly polluted conditions, such as exposure to smoke from burning materials or incense sticks, the sensor readings increased significantly. The ESP8266 processed this data and compared it with the threshold value. Once the threshold was exceeded, the system activated the buzzer, providing an immediate alert. The web interface also displayed high pollution levels, indicating hazardous conditions. This confirms that the system is capable of detecting dangerous environments and responding appropriately. The quick response time of the system ensures that users are alerted without delay, enhancing safety.

The real-time data transmission capability of the system was also analyzed. The ESP8266 successfully connected to the WiFi network and hosted a web server. The generated IP address allowed users to access the air quality data through a browser. The web interface displayed updated readings continuously, without noticeable delay. Multiple devices were used to access the system simultaneously, and the performance remained stable. This indicates that the communication module is efficient and capable of handling real-time data transmission.

The accuracy of the MQ-135 sensor was evaluated based on its response to environmental changes. While the sensor provided consistent and reliable readings, it was observed that external factors such as temperature, humidity, and calibration could influence its performance. The sensor is suitable for detecting relative changes in air quality rather than providing precise measurements. Despite this limitation, the system is effective for general monitoring purposes and can be improved through proper calibration and the use of advanced sensors in future enhancements.

The reliability of the system was assessed through continuous operation over extended periods. The system maintained stable performance without interruptions, indicating that it is suitable for long-term use. The ESP8266 handled multiple tasks, including sensor data processing, WiFi communication, and web server operation, efficiently. The hardware components showed no significant issues during testing, and the system operated smoothly under different conditions. This confirms the robustness of the system.

The user interface was analyzed based on its usability and accessibility. The web-based interface provided a clear and simple display of air quality data, making it easy for users to understand the information. The use of a browser-based interface eliminated the need for additional software, improving convenience. Users were able to access the system from different devices, enhancing flexibility. The interface design contributed to a positive user experience and ensured effective communication of data.

Despite the overall positive performance, certain limitations were identified during the analysis. The MQ-135 sensor requires calibration for improved accuracy, and its readings may vary due to environmental factors. The system depends on a stable WiFi connection, which may affect performance in areas with poor network coverage. Additionally, the system provides general air quality information and does not distinguish between specific gases. These limitations highlight areas for improvement in future development

CHAPTER VI

SYSTEM TESTING AND IMPLEMENTATION

6.1 SYSTEM TESTING

1. Introduction to System Testing

System testing is an essential phase in the development of any project, as it ensures that the system functions correctly and meets the intended objectives. In this project, system testing is performed to verify the performance, accuracy, reliability, and functionality of the IoT-based air quality monitoring device. The testing process involves evaluating both hardware and software components under different environmental conditions. The main goal of system testing is to identify errors, validate system behavior, and ensure that all modules work together seamlessly. It also helps in improving the overall quality and efficiency of the system before final deployment.

2. Hardware Testing

Hardware testing focuses on verifying the proper functioning of all physical components used in the system. The ESP8266 microcontroller, MQ-135 gas sensor, buzzer, and power supply are individually tested to ensure correct operation. The MQ-135 sensor is tested by exposing it to different environmental conditions to observe its response to varying gas concentrations. The ESP8266 is tested for proper data processing and WiFi connectivity. The buzzer is checked to ensure that it activates when the air quality exceeds the predefined threshold. The connections between components are also verified to ensure there are no loose or faulty connections. Hardware testing ensures that all components are functioning correctly and are properly integrated.

3. Software Testing

Software testing is carried out to verify the correctness and efficiency of the program written for the ESP8266 microcontroller. The code is tested in the Arduino IDE to ensure that there are no syntax or logical errors. Each function, such as sensor data reading, data processing, WiFi connection, web server setup, and buzzer control, is tested individually. The system is also tested for continuous operation to ensure that it can handle real-time data processing without failure. Debugging techniques are used to identify and fix errors in the program. Software testing ensures that the system performs all operations accurately and efficiently.

4.Functional Testing

Functional testing is performed to verify that the system operates according to the specified requirements. The system is tested to ensure that it can detect air quality, process data, display results on the web interface, and activate the buzzer when necessary. Different scenarios are created to test the system's functionality. For example, the sensor is exposed to clean air to check if the system identifies safe conditions, and then exposed to polluted air to verify if it detects hazardous conditions and triggers the alert. The results confirm that the system performs all intended functions correctly.

5. Connectivity Testing

Connectivity testing is conducted to evaluate the WiFi communication and data transmission capabilities of the system. The ESP8266 is tested for its ability to connect to a local WiFi network and generate an IP address. The web interface is accessed using different devices such as smartphones, laptops, and tablets to ensure compatibility. The system is tested for real-time data updates and response to multiple users accessing the data simultaneously. The results show that the system maintains stable connectivity and provides efficient data transmission without significant delays.

6. Performance Testing

Performance testing evaluates the efficiency and responsiveness of the system under different conditions. The system is tested for response time, accuracy, and continuous operation. The MQ-135 sensor is observed for its response to sudden changes in air quality, and the ESP8266 is evaluated for its ability to process data quickly. The system is also tested for long-term operation to ensure stability. The results indicate that the system performs efficiently, with quick response times and consistent output.

7. Validation and Result Analysis

After completing all testing procedures, the system is validated to ensure that it meets the project objectives. The results obtained during testing are analyzed to verify accuracy and reliability. The system successfully detects changes in air quality, provides real-time data, and generates alerts when necessary. Minor variations in sensor readings are observed due to environmental factors, but overall performance remains satisfactory. The validation process confirms that the system is suitable for practical applications.

6.1.1 UNIT TESTING

1. Introduction to Unit Testing

Unit testing is the process of testing individual components or modules of a system separately to ensure that each unit functions correctly. In the proposed air quality monitoring system, unit testing plays a crucial role in verifying the performance of each hardware and software module before integrating them into a complete system. The main objective of unit testing is to detect errors at an early stage and ensure that each module operates according to its design specifications. By testing each unit independently, the overall reliability and efficiency of the system can be improved significantly.

2. Sensor Module Testing

The MQ-135 gas sensor is tested individually to evaluate its ability to detect different gas concentrations. During testing, the sensor is exposed to various environmental conditions such as clean air, smoke, and polluted air sources. The analog output from the sensor is monitored using the serial monitor in the Arduino IDE. The variation in sensor readings confirms that the sensor is responsive to changes in air quality. Calibration is also performed during this stage to improve the accuracy of the sensor. Any inconsistencies in readings are identified and corrected to ensure reliable performance.

3. Microcontroller Unit Testing

The ESP8266 microcontroller is tested separately to verify its processing and control capabilities. The testing includes checking its ability to read sensor data, execute programmed instructions, and control connected components such as the buzzer. Simple test programs are uploaded to ensure that the microcontroller is functioning properly. The serial monitor is used to display output values, which helps in verifying correct data processing. The microcontroller is also tested for stability and continuous operation, ensuring that it can handle real-time data processing without errors.

4. WiFi Module Testing

The WiFi functionality of the ESP8266 is tested to ensure proper connectivity and communication. The module is programmed to connect to a local WiFi network using the SSID and password. Once connected, the IP address generated by the ESP8266 is displayed on the serial monitor. This IP address is used to access the web interface. The connection is tested multiple times to ensure stability and reliability. The ability of the module to reconnect automatically after disconnection is also verified. This testing ensures that the communication module works efficiently.

5. Web Server and Interface Testing

The web server hosted by the ESP8266 is tested to verify its ability to display air quality data. The HTML code embedded in the program is checked for correctness and proper formatting. The web page is accessed through different devices such as smartphones, laptops, and tablets to ensure compatibility. The real-time updating of sensor data on the web page is observed and verified. The interface is tested for responsiveness and clarity, ensuring that users can easily understand the displayed information. This testing confirms that the display module functions correctly.

6. Buzzer (Alert Module) Testing

The buzzer is tested independently to verify its functionality as an alert system. The ESP8266 is programmed to activate the buzzer when a certain condition is met, such as exceeding a predefined threshold value. During testing, different conditions are simulated to check whether the buzzer responds correctly. The buzzer is activated manually and automatically to ensure proper operation. The sound intensity and response time are also evaluated. This testing ensures that the alert mechanism works effectively in real-time situations.

7. Power Supply Testing

The power supply unit is tested to ensure that it provides a stable and consistent voltage to all components. The system is powered using a 5V DC supply through a USB connection or external adapter. The voltage levels are monitored to ensure that there are no fluctuations that could affect system performance. The components are tested under continuous operation to verify that the power supply supports long-term usage. This testing ensures the reliability and stability of the system.

8. Integration Readiness Verification

After testing each unit individually, the system is evaluated for integration readiness. This step ensures that all modules are functioning correctly and are compatible with each other. Any issues identified during unit testing are resolved before moving to system integration. This reduces the chances of errors during the final implementation.

The successful completion of unit testing indicates that the system components are ready to be combined into a complete working system.

6.1.2 INTEGRATION TESTING

1. Introduction to Integration Testing

Integration testing is the process of combining individual modules of a system and testing them as a group to ensure that they work together correctly. After completing unit testing for each component, integration testing is performed to verify the interaction between hardware and software modules in the proposed air quality monitoring system. The main objective of this phase is to identify issues related to data flow, communication, and coordination between different modules. It ensures that the complete system functions smoothly when all components are connected and operating together.

2. Sensor and Microcontroller Integration

The first step in integration testing involves connecting the MQ-135 gas sensor with the ESP8266 microcontroller. The sensor output is linked to the microcontroller input, allowing real-time data transfer. During testing, the sensor readings are observed through the serial monitor to ensure that the ESP8266 correctly receives and processes the data. The system is tested under different environmental conditions to verify that the sensor values are accurately captured and interpreted by the microcontroller. This integration confirms that the sensing and processing modules work effectively together.

3. Microcontroller and WiFi Communication Integration

The next stage of integration testing focuses on the communication between the ESP8266 microcontroller and the WiFi network. The microcontroller is programmed to connect to a local network and host a web server. Once connected, the IP address generated by the ESP8266 is used to access the system through a web browser. During testing, the system is checked for stable connectivity and proper data transmission. The ability of the ESP8266 to send processed data to the web interface in real time is verified. This ensures that the communication module is properly integrated with the processing unit.

4. Web Interface Integration

The integration of the web interface with the ESP8266 is tested to ensure that the system can display real-time air quality data. The web page hosted by the ESP8266 is accessed through multiple devices, such as smartphones and laptops. The data displayed on the web page is compared with the sensor readings observed on the serial monitor to verify accuracy. The interface is tested for responsiveness and continuous updates. This confirms that the display module is successfully integrated with the system and provides accurate information to the user.

5. Alert System Integration

The buzzer is integrated with the ESP8266 to provide an alert mechanism. During testing, the system is exposed to conditions where the air quality exceeds the predefined threshold. The ESP8266 processes the sensor data and activates the buzzer when the threshold is crossed. The response time and accuracy of the alert system are observed to ensure proper functioning. The buzzer remains inactive during safe conditions, confirming that it responds only when required. This integration ensures that the alert module works in coordination with the sensor and processing modules.

6. End-to-End System Integration

The final stage of integration testing involves testing the complete system as a whole. All modules, including the sensor, microcontroller, WiFi communication, web interface, and alert system, are connected and tested together. The system is observed for continuous operation, real-time data updates, and proper alert generation. Different environmental conditions are simulated to evaluate system performance. The results confirm that the system operates smoothly, with all components working together effectively.

6.1.3 SYSTEM IMPLEMENTATION AND MAINTENANCE

1. Introduction to Implementation and Maintenance

System implementation and maintenance are essential phases in the lifecycle of the IoT-based air quality monitoring system. Implementation refers to the process of deploying the developed system into a real-world environment, while maintenance ensures that the system continues to function efficiently over time. These phases are critical to achieving reliable performance, long-term usability, and user satisfaction. The implementation process involves setting up the hardware, configuring the software, and validating system functionality, whereas maintenance focuses on monitoring, updating, and troubleshooting the system after deployment.

2. System Implementation Process

The implementation of the air quality monitoring system begins with the preparation of hardware components such as the ESP8266 microcontroller, MQ-135 gas sensor, buzzer, and supporting elements like breadboard and jumper wires. The components are assembled carefully according to the circuit design to ensure proper connectivity. Once the hardware setup is complete, the software is developed and uploaded to the ESP8266 using the Arduino IDE. The program includes instructions for sensor data acquisition, WiFi connectivity, web server operation, and alert generation.

After programming, the system is powered on, and the ESP8266 attempts to connect to the configured WiFi network. Once connected, it generates an IP address, which is used to access the web interface. The implementation process also includes verifying the correct functioning of all modules by observing sensor readings, checking data display on the web page, and testing the buzzer for alert conditions. This step ensures that the system is fully operational before deployment.

3. Deployment of the System

The deployment phase involves placing the system in the desired environment where air quality needs to be monitored. The device can be installed in locations such as homes, offices, laboratories, or industrial areas. Proper placement of the MQ-135 sensor is important to ensure accurate readings, as it should be exposed to ambient air without obstruction. The system is powered using a stable 5V supply and connected to a reliable WiFi network for continuous operation. Once deployed, the system begins monitoring air quality in real time and provides data through the web interface.

4. Maintenance of Hardware Components

Maintenance of hardware components is necessary to ensure the long-term reliability of the system. The MQ-135 sensor may accumulate dust or contaminants over time, which can affect its accuracy. Regular cleaning and recalibration of the sensor are recommended to maintain performance. The connections between components should be checked periodically to ensure there are no loose wires or faults. The power supply should also be monitored to avoid voltage fluctuations that could damage the components. Proper hardware maintenance helps in extending the lifespan of the system.

5. Software Maintenance and Updates

Software maintenance involves updating the system code to improve performance, fix bugs, and add new features. The Arduino program can be modified to enhance data processing, improve the user interface, or integrate additional functionalities. Periodic updates ensure that the system remains efficient and compatible with evolving technologies. Backup of the source code is also important to prevent data loss. Software maintenance plays a key role in keeping the system up-to-date and functional.

6. System Monitoring and Troubleshooting

Continuous monitoring of the system is essential to detect and resolve issues promptly. Users should regularly check the web interface to ensure that data is being displayed correctly. In case of any issues, basic troubleshooting steps can be followed, such as checking the power supply, verifying WiFi connectivity, and restarting the system. If the sensor readings appear inaccurate, recalibration may be required. Proper troubleshooting ensures that the system operates smoothly without major interruptions.

7. User Support and Documentation

Providing user support and proper documentation is an important aspect of system maintenance. A user manual should be available with detailed instructions on system setup, operation, and troubleshooting. This helps users understand how to use the system effectively and resolve minor issues independently. Documentation also includes information about hardware components, software requirements, and maintenance procedures. Good documentation enhances user experience and reduces dependency on technical support.

6.2 SYSTEM IMPLEMETATION

6.2.1 IMPLEMENTATION PLAN

The implementation plan defines the structured approach followed to deploy the IoT-based air quality monitoring system in a real-world environment. It includes a sequence of steps starting from system preparation to final deployment and monitoring. The first stage of the implementation plan involves requirement analysis and preparation, where all necessary hardware components such as the ESP8266 microcontroller, MQ-135 gas sensor, buzzer, and supporting components are collected and verified. The software environment is also prepared by installing the Arduino IDE and required libraries for programming and communication.

The next stage involves system setup and configuration. The hardware components are assembled using a breadboard and jumper wires, ensuring proper connections between the sensor, microcontroller, and buzzer. The ESP8266 is programmed using Arduino IDE with the required code for sensor data acquisition, WiFi connectivity, web server hosting, and alert generation. The WiFi credentials are configured in the program to establish a network connection. After uploading the code, the system is powered on, and the serial monitor is used to verify successful initialization and network connection.

Following setup, the system undergoes testing and validation. The device is tested under different environmental conditions to ensure accurate data collection and proper functioning of all modules. The web interface is accessed using the IP address generated by the ESP8266 to verify real-time data display. The buzzer is tested by simulating high pollution conditions to confirm alert functionality. Any errors identified during testing are corrected, and the system is fine-tuned for optimal performance.

The final stage of the implementation plan is deployment and monitoring. The system is installed in the desired location, such as a home, office, or industrial environment. The device continuously monitors air quality and provides real-time data through the web interface. Regular monitoring ensures that the system operates effectively and provides accurate information. The implementation plan ensures a smooth transition from development to real-world application, minimizing errors and ensuring reliable performance.

6.2.2 USER TRAINING AND DOCUMENTATION

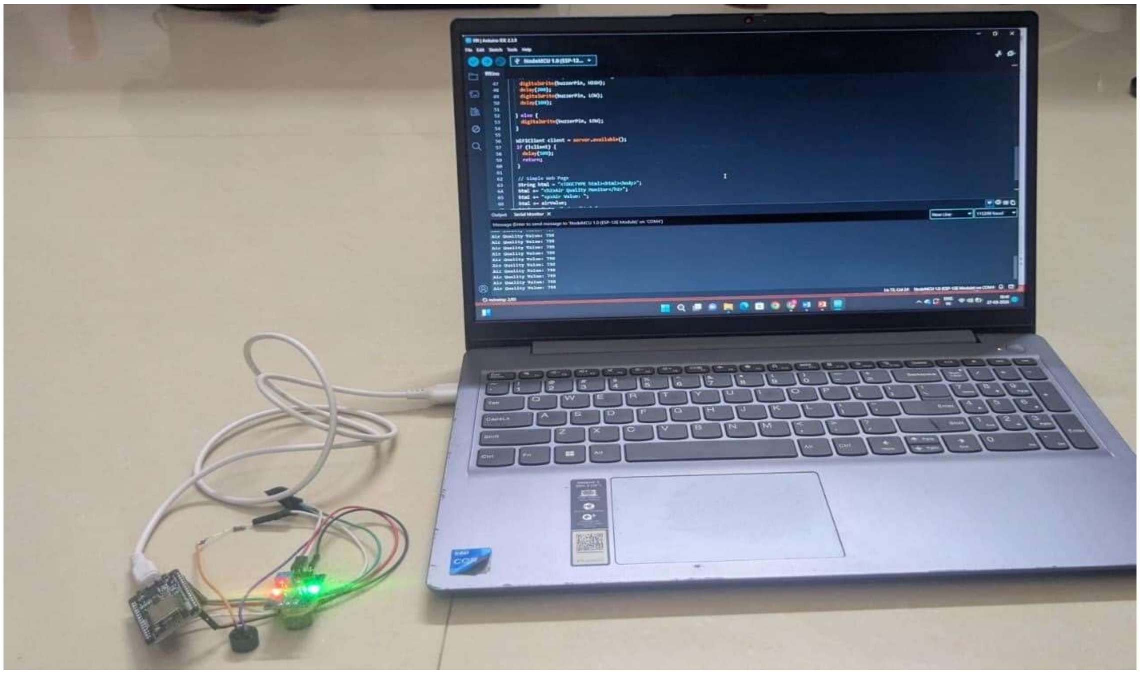
User training is an important aspect of system implementation, as it ensures that users can effectively operate and maintain the air quality monitoring system. The training process includes educating users about the basic functionality of the system, including how to power the device, connect it to a WiFi network, and access the web interface. Users are guided on how to enter the IP address into a web browser to view real-time air quality data. They are also trained to understand the displayed values and interpret whether the air quality is safe or hazardous.

In addition to basic operation, users are trained on how to respond to alerts generated by the system. When the buzzer is activated, users should take appropriate actions such as improving ventilation or avoiding exposure to polluted environments. Training also includes simple troubleshooting techniques, such as checking power supply connections, verifying WiFi connectivity, and restarting the system if necessary. This ensures that users can handle minor issues without technical assistance.

Documentation plays a key role in supporting user training and system maintenance. A user manual is provided, containing detailed instructions on system setup, operation, and troubleshooting. The documentation includes information about hardware components, software requirements, and step-by-step procedures for installation and usage. Diagrams and screenshots are included to make the instructions clear and easy to follow. Proper documentation ensures that users can refer to the guidelines whenever needed, reducing dependency on external support.

Maintenance guidelines are also included in the documentation to ensure long-term system performance. Users are advised to periodically check the sensor for dust or contamination, which may affect accuracy. The system should be recalibrated if necessary to maintain reliable readings. Regular updates to the software can also improve system performance and add new features. Proper maintenance ensures that the system continues to function efficiently over time.

6.3 SYSTEM PROTOTYPE:



CHAPTER VII

CONCLUSION AND FUTURE ENHANCEMENT

7.1 CONCLUSION

The wearable air quality monitoring watch developed in this project provides an effective solution for real-time detection of environmental pollution. By using the MQ-135 gas sensor and NodeMCU microcontroller, the system is able to measure the concentration of harmful gases and display the air quality in PPM values along with status indicators such as good, moderate, or poor.

The device is compact, portable, and easy to use, making it suitable for daily life applications. It helps users stay aware of their surrounding air quality and take necessary precautions when pollution levels rise. The inclusion of an alert system further enhances safety by notifying users when air quality becomes unsafe.

The system was successfully tested for accuracy, performance, and reliability, ensuring that it meets the required functional and usability standards. Overall, this project demonstrates how simple and low-cost technology can be used to address real-world environmental problems and promote health awareness.

7.2 SCOPE FOR FUTURE ENHANCEMENT

Although the wearable air quality monitoring watch is efficient and functional, there are several possible improvements that can enhance its performance and usability in the future.

Future Enhancements:

Integration of Advanced Sensors

One of the major future enhancements of the proposed system is the integration of advanced and high-precision sensors. Currently, the system uses the MQ-135 gas sensor, which provides general air quality detection but has limitations in accuracy and specificity. In future developments, more advanced sensors such as particulate matter sensors (PM2.5 and PM10), carbon monoxide (CO), nitrogen dioxide (NO_2), and ozone (O_3) sensors can be incorporated. These sensors can provide detailed information about specific pollutants present in the environment. The addition of such sensors will improve the accuracy and reliability of the system, making it suitable for professional and industrial applications. This enhancement will also allow the system to comply with environmental standards and provide more meaningful data for analysis.

Cloud Integration and Data Storage

The current system displays real-time data through a web interface but does not store historical data. In the future, the system can be enhanced by integrating cloud platforms such as AWS, Google Cloud, or ThingSpeak. This will enable the storage of air quality data over long periods, allowing users to analyze trends and patterns. Cloud integration will also support remote monitoring from anywhere in the world, not just within a local network. Historical data can be used for research, environmental studies, and decision-making. This enhancement will transform the system from a basic monitoring device into a comprehensive data analysis platform.

Mobile Application Development

Another important enhancement is the development of a dedicated mobile application for the system. While the current system uses a web-based interface, a mobile app can provide a more interactive and user-friendly experience. The app can display real-time air quality data, send notifications, and provide alerts when pollution levels exceed safe limits. It can also include graphical representations such as charts and graphs for better visualization of data. Mobile applications can improve accessibility and allow users to monitor air quality conveniently from their smartphones. This enhancement will increase user engagement and make the system more practical for everyday use.

Integration of GPS Module

Adding a GPS module to the system can enable location-based air quality monitoring. This feature is particularly useful for mobile applications, where the device can be carried to different locations. The system can record air quality data along with geographic coordinates, allowing users to map pollution levels across different areas. This enhancement can be used for environmental surveys, urban planning, and research purposes. It can also help in identifying pollution hotspots and taking appropriate measures to control them.

AI and Machine Learning Integration

The integration of Artificial Intelligence (AI) and Machine Learning (ML) can significantly enhance the capabilities of the system. AI algorithms can be used to analyze historical data and predict future air quality trends. Machine learning models can identify patterns in pollution levels and provide insights into environmental conditions. This can help users take preventive actions before pollution levels become hazardous. AI-based systems can also improve sensor calibration and accuracy by learning from past data. This enhancement will make the system more intelligent and capable of advanced data analysis.

Multi-Parameter Environmental Monitoring

Currently, the system focuses only on air quality monitoring. In the future, it can be expanded to monitor multiple environmental parameters such as temperature, humidity, noise levels, and light intensity. By integrating additional sensors, the system can provide a complete overview of environmental conditions. This multi-parameter monitoring system can be used in smart homes, smart cities, and industrial environments. It will provide comprehensive data that can be used for better decision-making and environmental management.

Improved Power Management and Battery Support

Power management is an important aspect of IoT devices. Future enhancements can include the use of rechargeable batteries and energy-efficient components to improve system performance. Solar power integration can also be considered for outdoor applications, allowing the system to operate independently without relying on external power sources. Low-power modes can be implemented to reduce energy consumption during idle periods. These improvements will make the system more sustainable and suitable for long-term deployment.

Enhanced Alert and Notification System

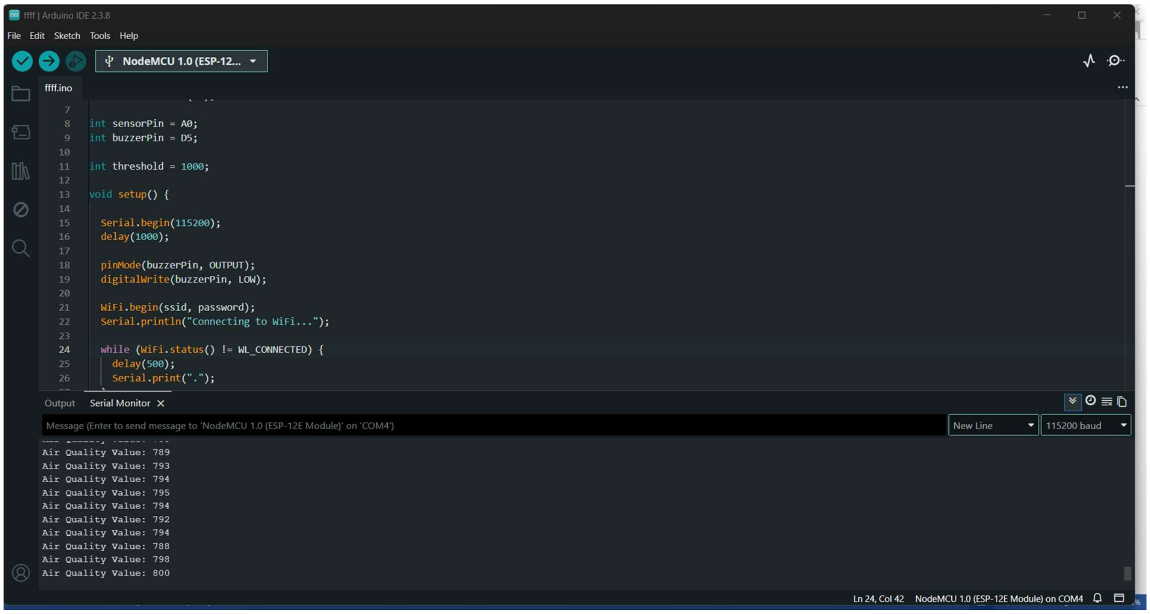
The current system uses a buzzer for alerts, which provides basic notification. In the future, the alert system can be enhanced by integrating SMS, email, and push notifications. This will ensure that users are notified even when they are not near the device. Notifications can include detailed information about air quality levels and recommended actions. This enhancement will improve user awareness and enable quicker responses to hazardous conditions.

Improved User Interface and Data Visualization

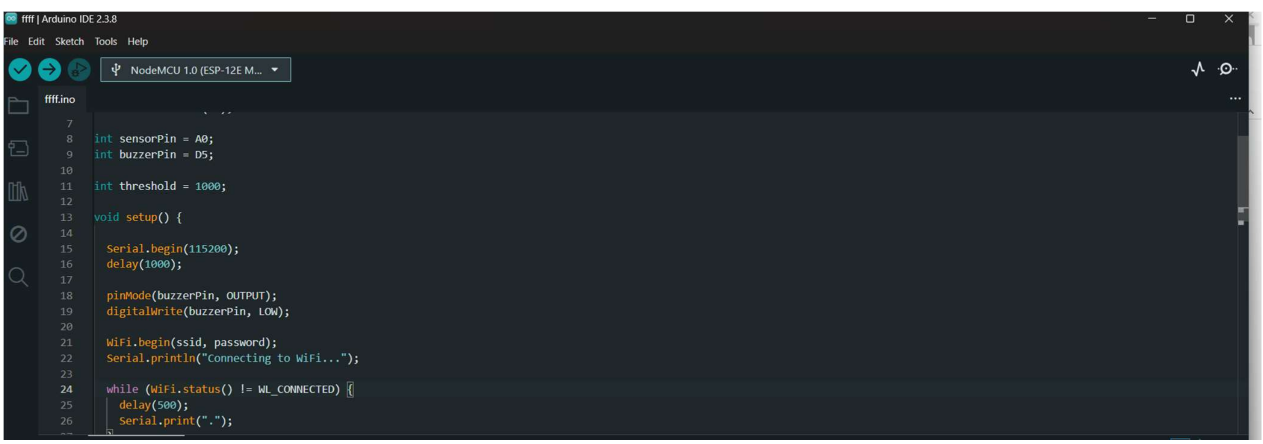
The web interface can be further enhanced to provide better visualization of air quality data. Advanced graphical elements such as charts, graphs, and dashboards can be added to make the data more interactive and easier to understand. Color-coded indicators can be used to represent different levels of air quality. These improvements will enhance the user experience and make the system more attractive and informative.

APPENDIX I

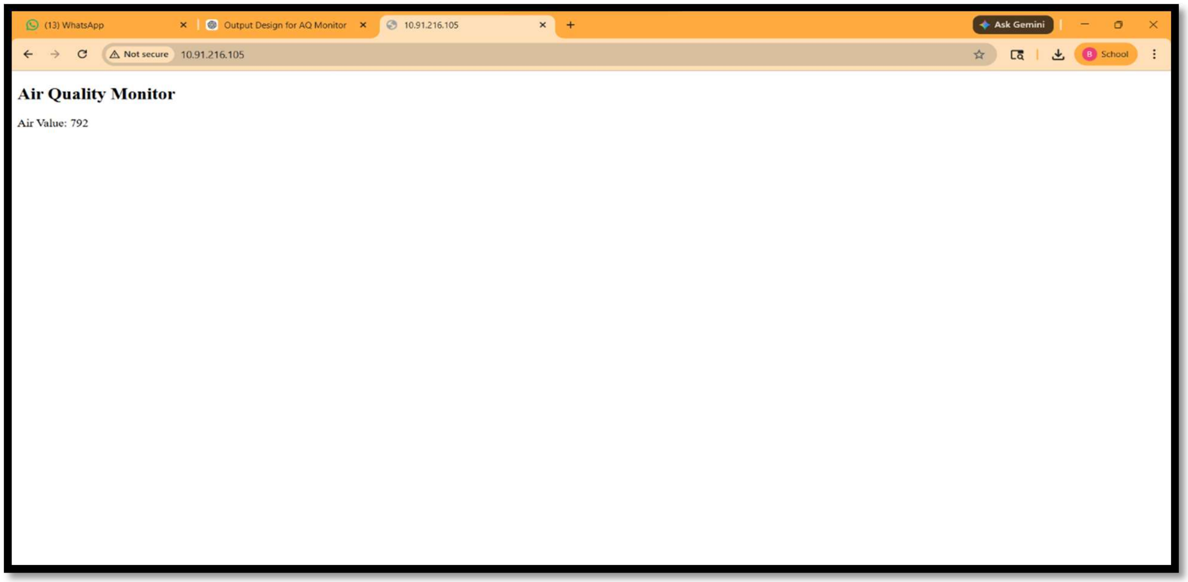
SCREENSHOT 1:



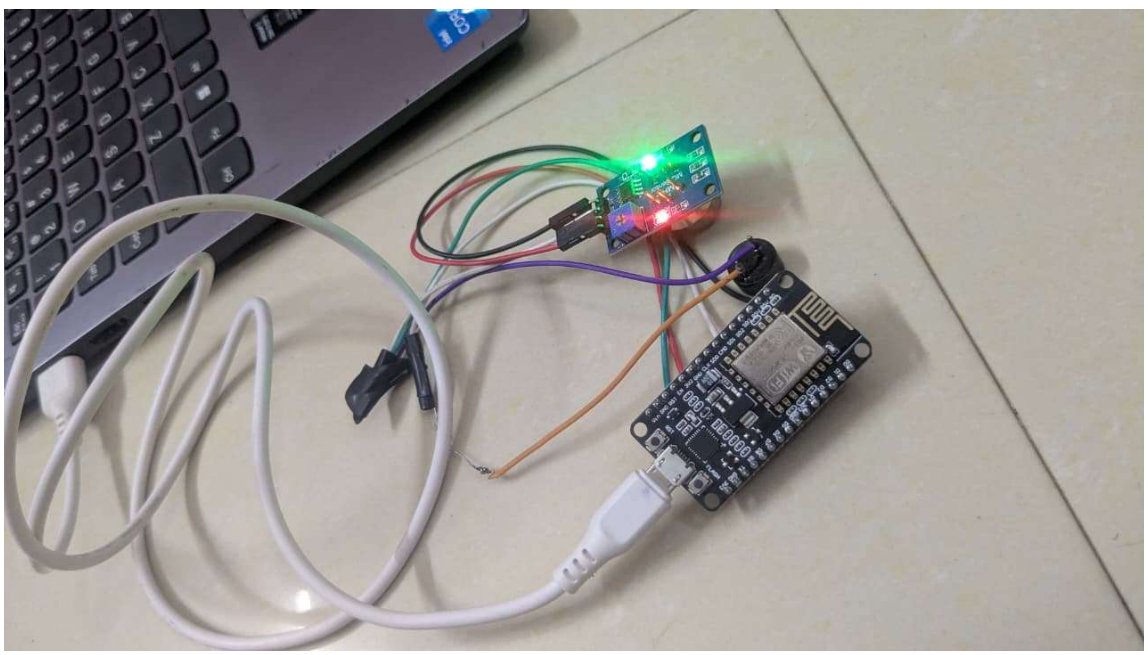
SCREENSHOT 2:



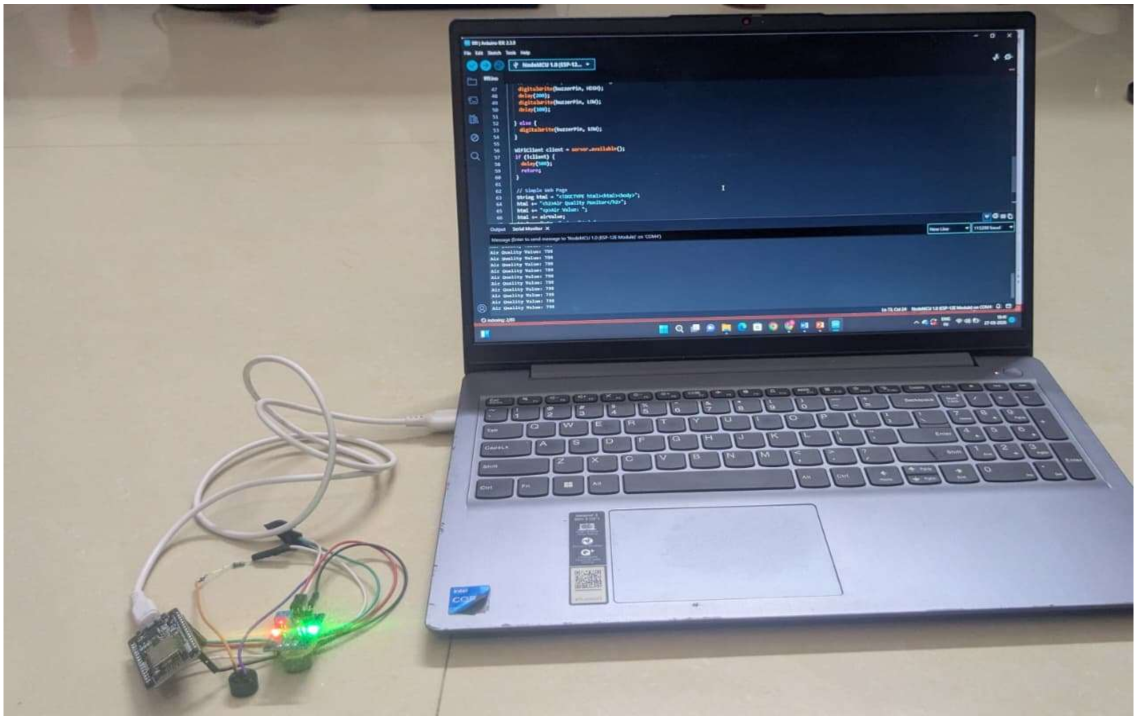
SCREENSHOT 3:



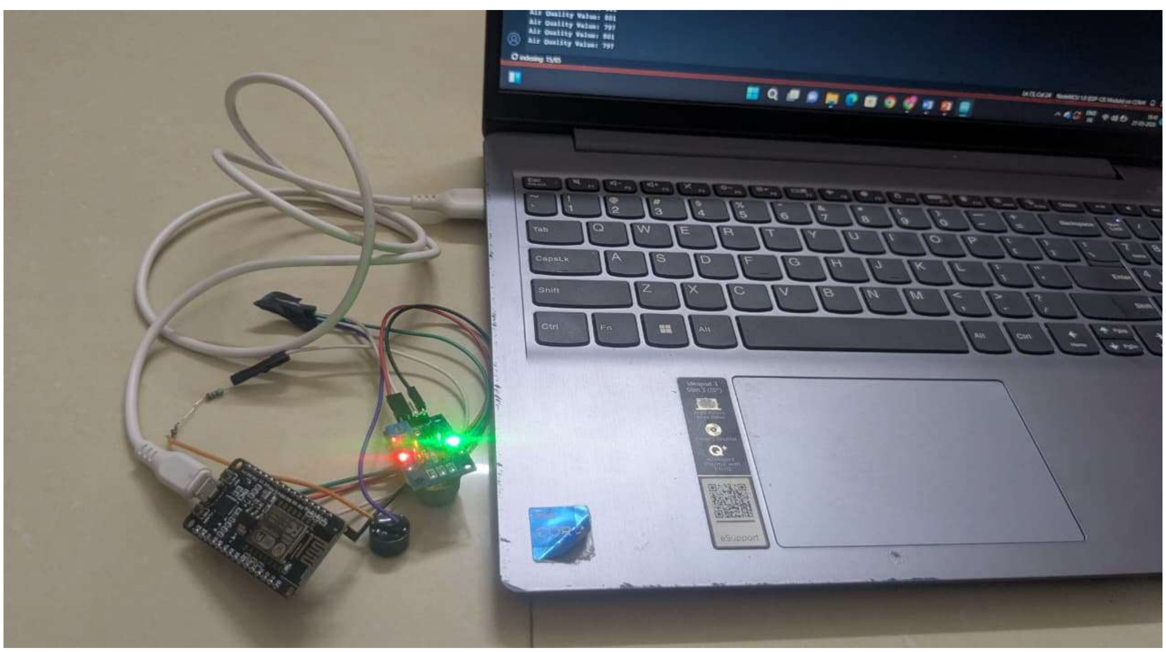
SCREENSHOT 4:



SCREENSHOT 5:



SCREENSHOT 6:



APPENDIX II

```
int sensorPin = A0;

 int buzzerPin = D5;


int threshold = 1000;

void setup() {

  Serial.begin(115200);   delay(1000);

  pinMode(buzzerPin, OUTPUT);   digitalWrite(buzzerPin, LOW);


  WiFi.begin(ssid, password);
  Serial.println("Connecting to WiFi...");

  while (WiFi.status() != WL_CONNECTED) {     delay(500);     Serial.print(".");
  }

  Serial.println("\nWiFi Connected!");
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());

  server.begin();
}

void loop() {

  int airValue = analogRead(sensorPin);

  Serial.print("Air Quality Value: ");
  Serial.println(airValue);

  // 🔊 LOUD ACTIVE BUZZER LOGIC
  if (airValue > threshold) {
```

```
   // Fast ON/OFF pattern for strong alert sound     digitalWrite(buzzerPin, HIGH);     delay(200);
digitalWrite(buzzerPin, LOW);     delay(100);


  } else {      digitalWrite(buzzerPin, LOW);
  }


  WiFiClient client = server.available();    if (!client) {     delay(500);     return;
  }


  // Simple Web Page
  String html = "<!DOCTYPE html><html><body>";   html += "<h2>Air Quality Monitor</h2>";
html +=
"<p>Air Value: ";   html += airValue;   html += "</p></body></html>";


  client.println("HTTP/1.1             200        OK");                          client.println("Content-Type:
           text/html");   client.println("Connection: close");   client.println();   client.println(html);


  delay(1);   client.stop();
}
```

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